

David L. Nelson and Michael M. Cox

LEHNINGER

PRINCIPLES OF BIOCHEMISTRY

Fifth Edition

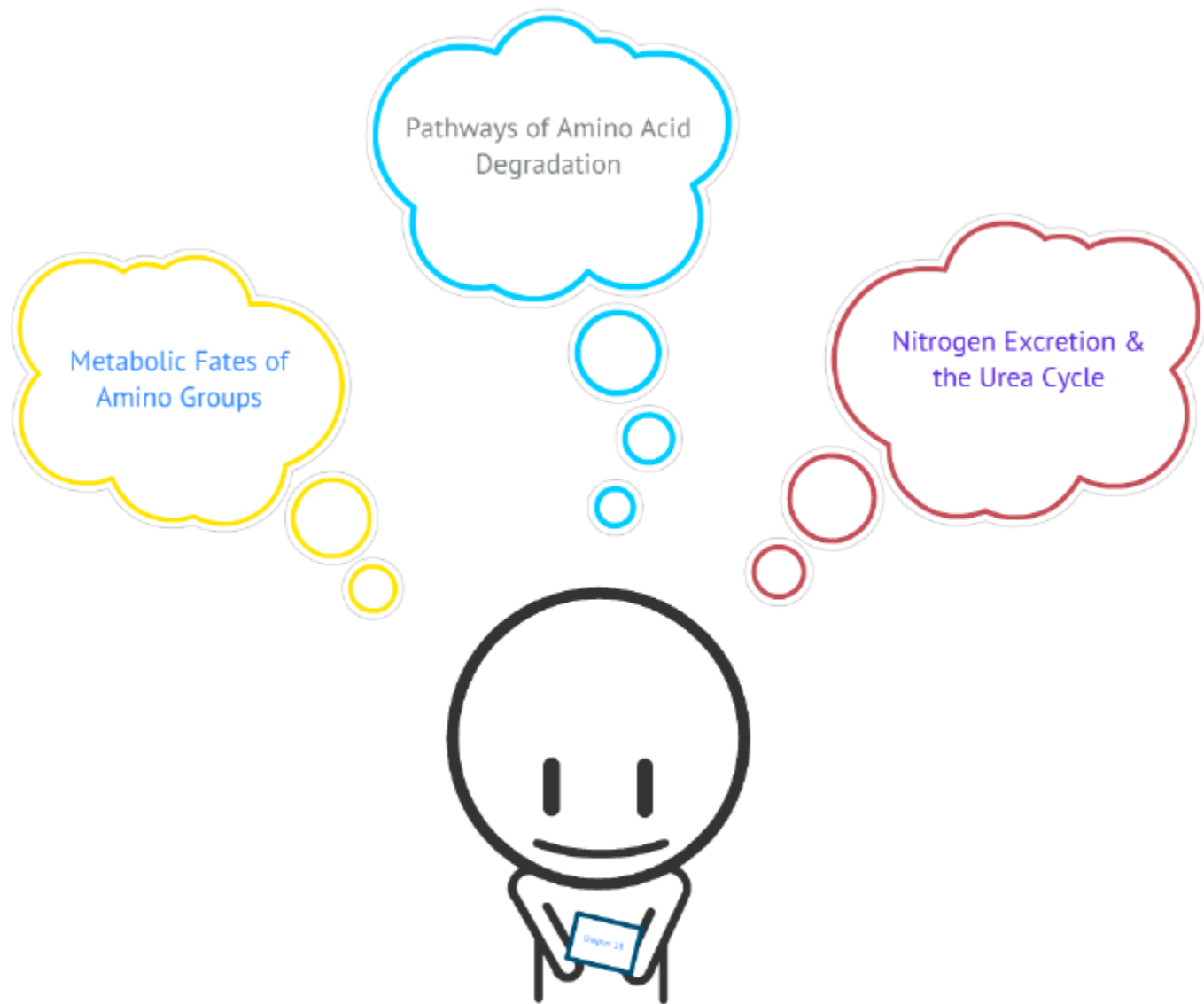
CHAPTER 18

**Amino Acid Oxidation and the
Production of Urea**

**Max hug
time
3 minutes**



**For fonder
farewells
please use
the car park**



Amino acid oxidation & Production of Urea

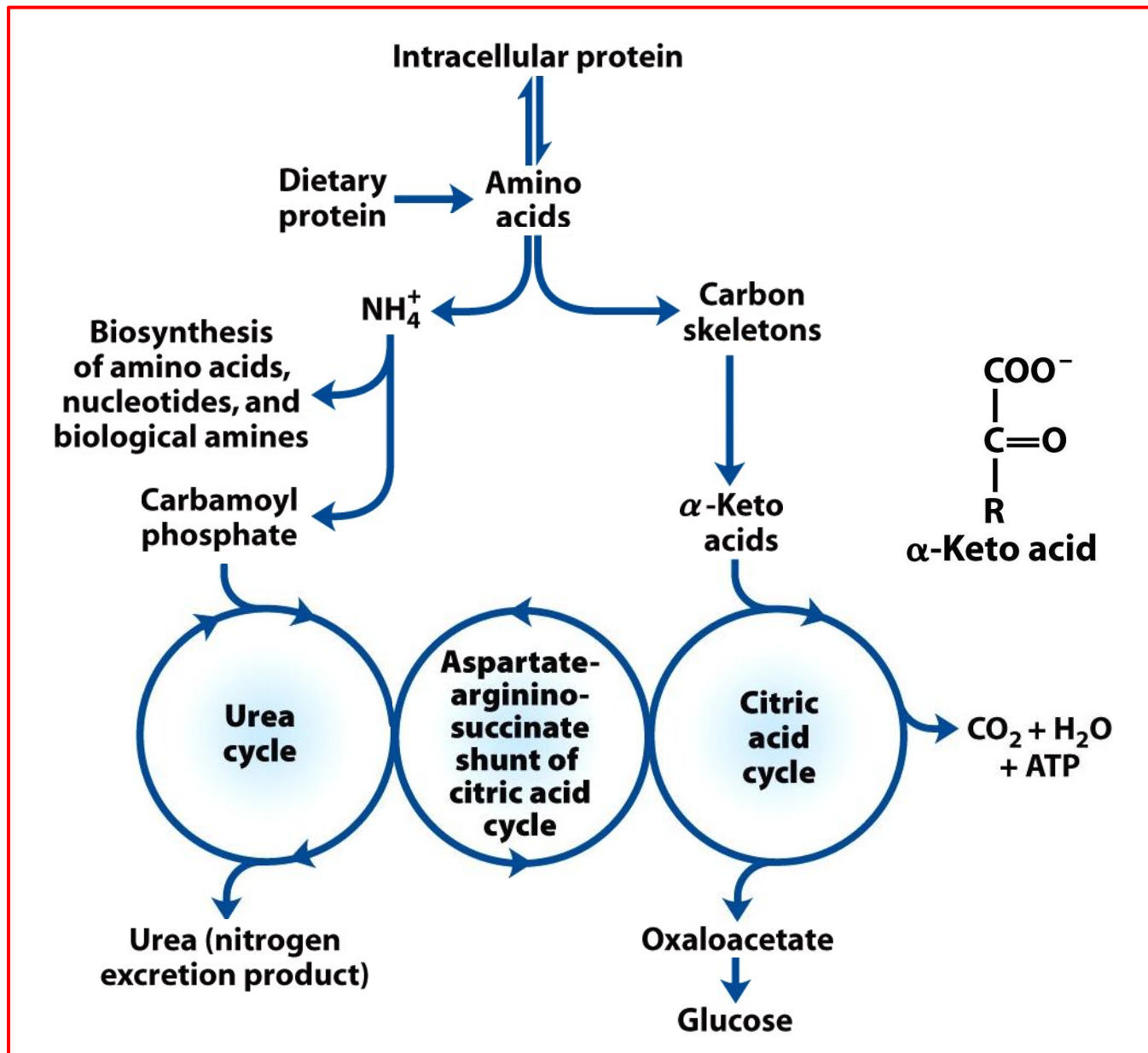
Amino acid metabolism

Important for carnivores
(obtain 90% of energy)

- 1.Synthesis & degradation of proteins
- 2.Dietary amino acids
- 3.Starvation or diabetes



Overview of amino acid catabolism in mammals

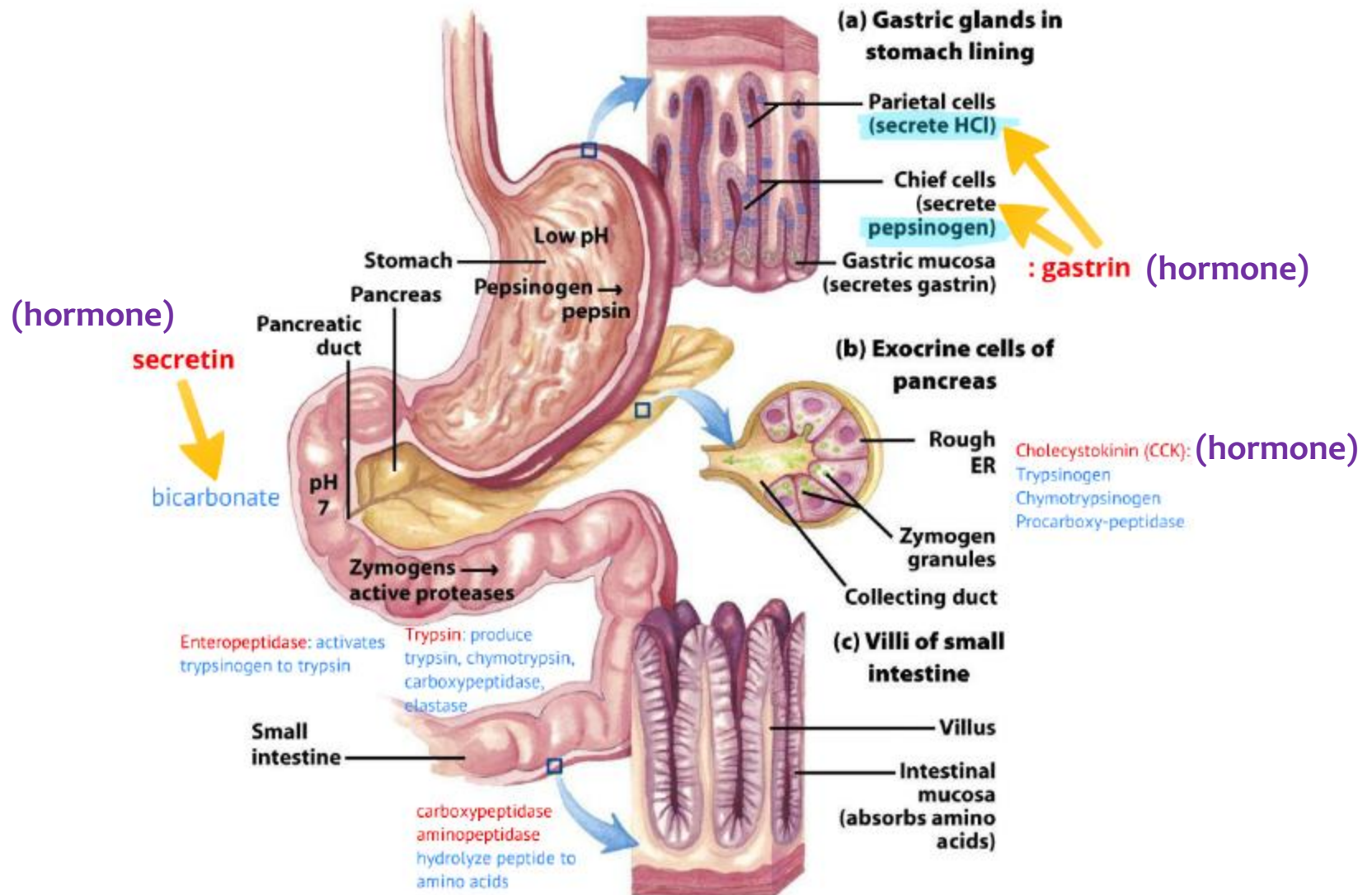


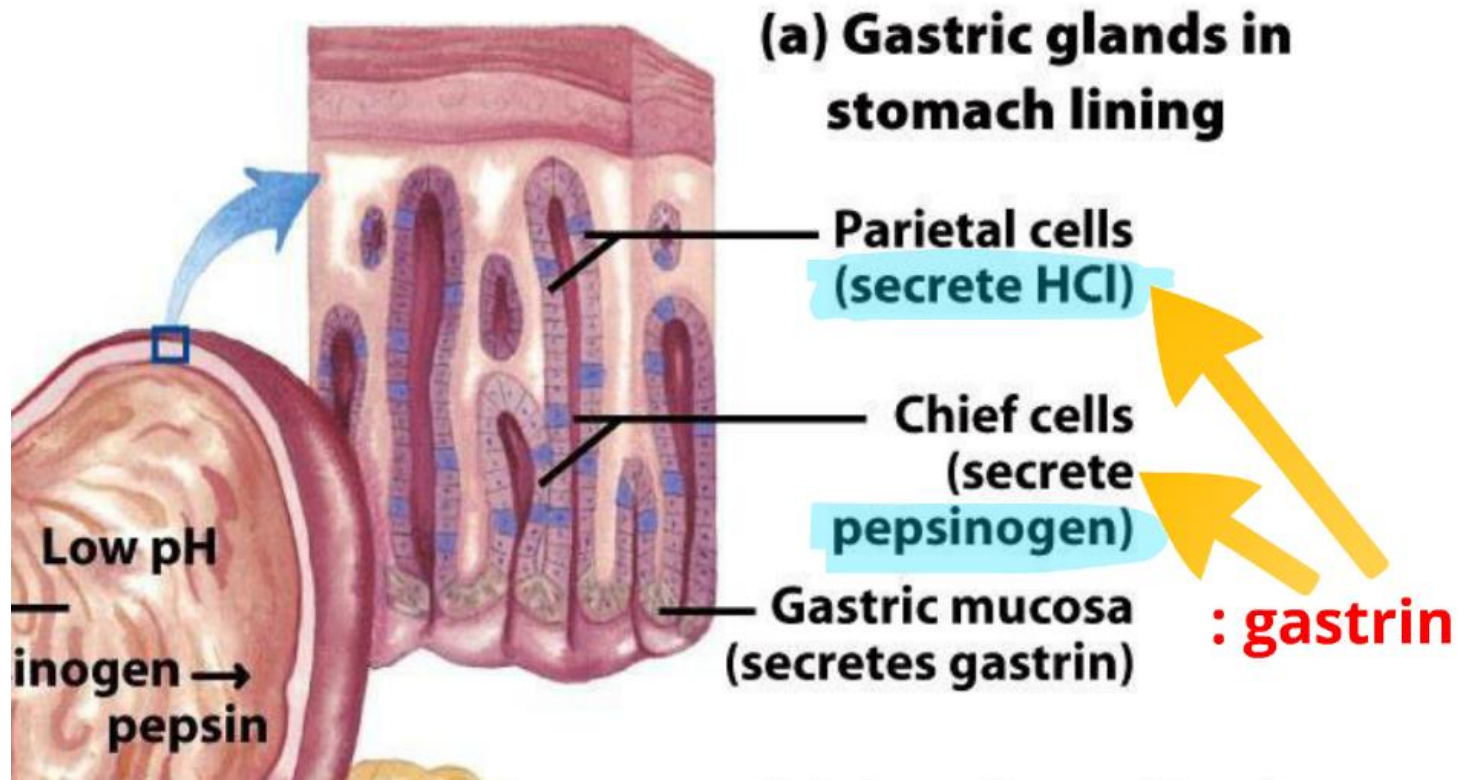
18.1 Metabolic fates of amino acids

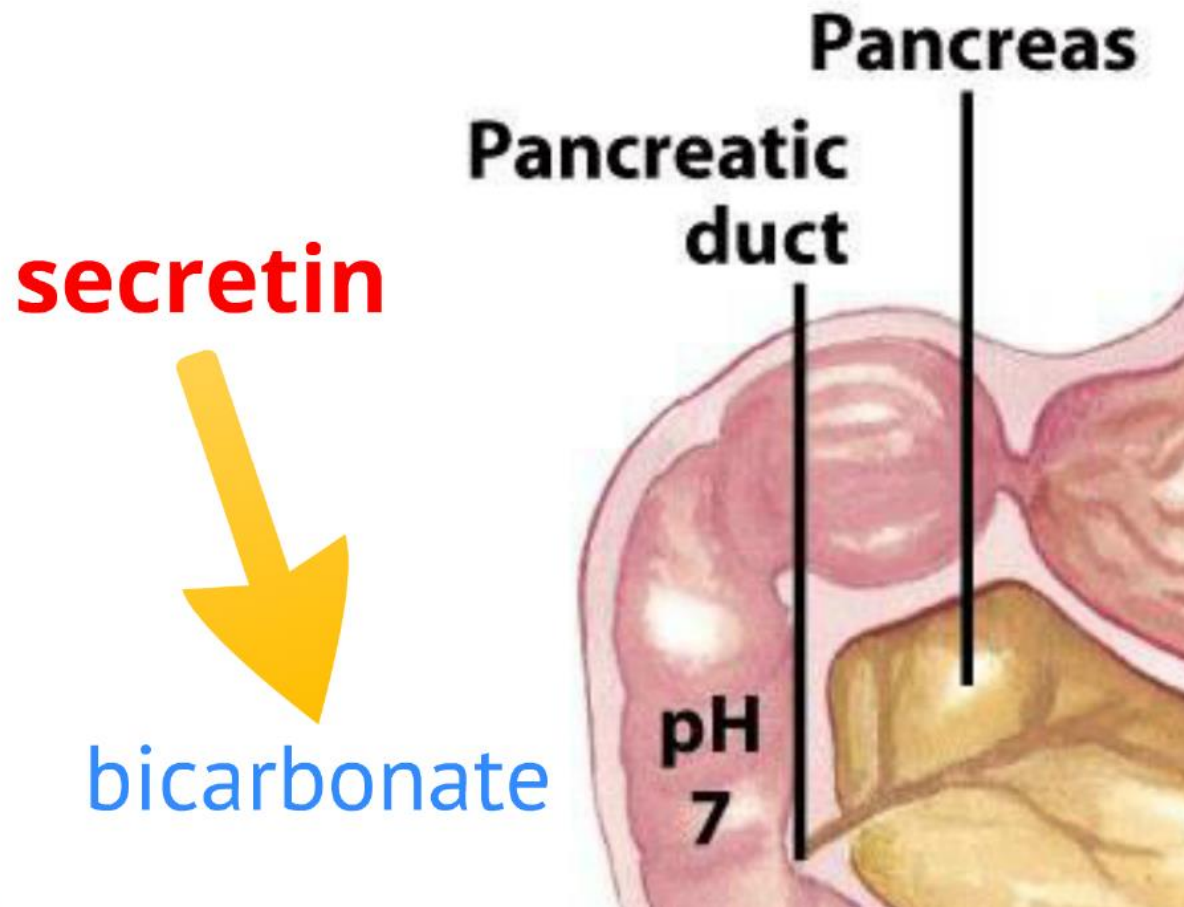


Metabolic Fates of
Amino Groups

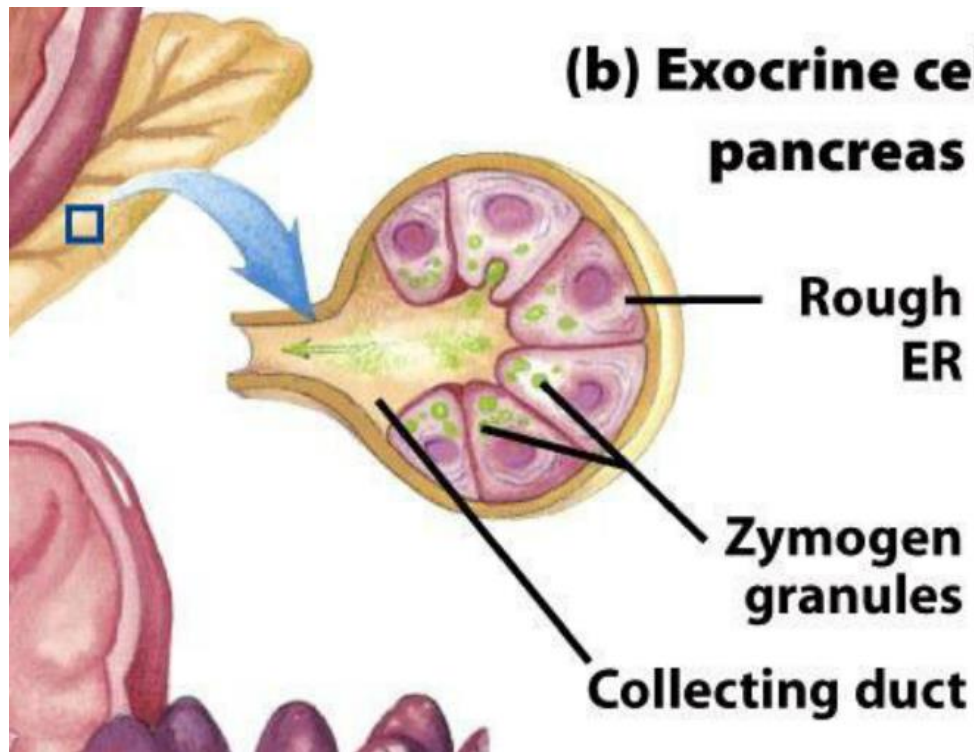
dietary protein is enzymatically degraded to amino acids







(b) Exocrine cells of pancreas

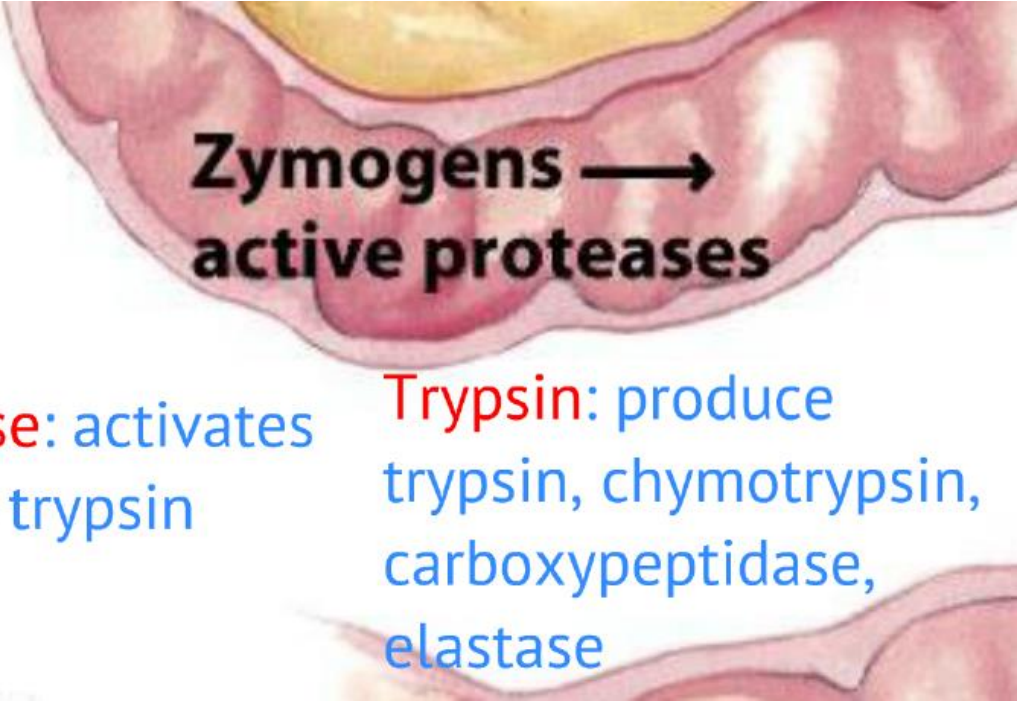


Cholecystokinin (CCK):

Trypsinogen

Chymotrypsinogen

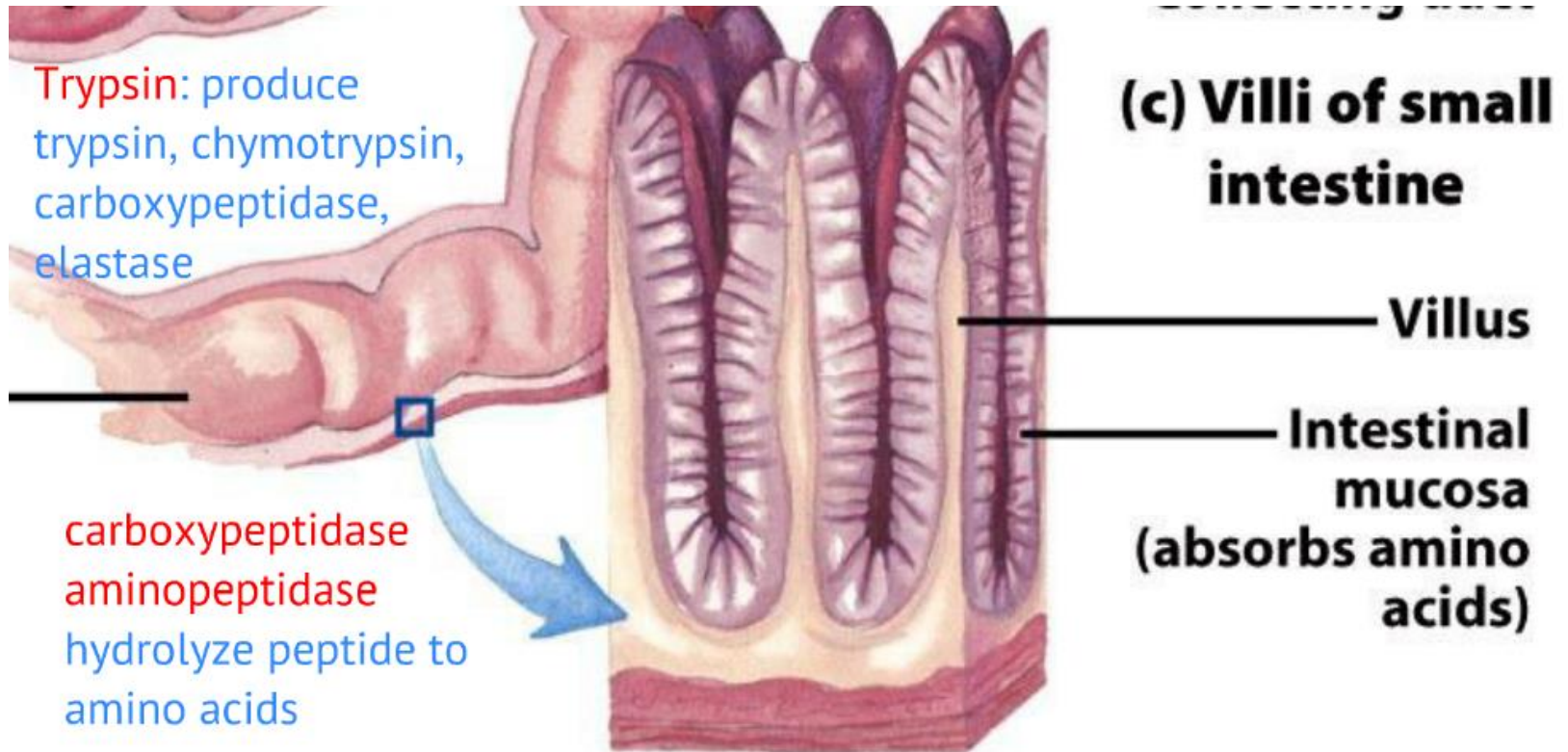
Procarboxy-peptidase



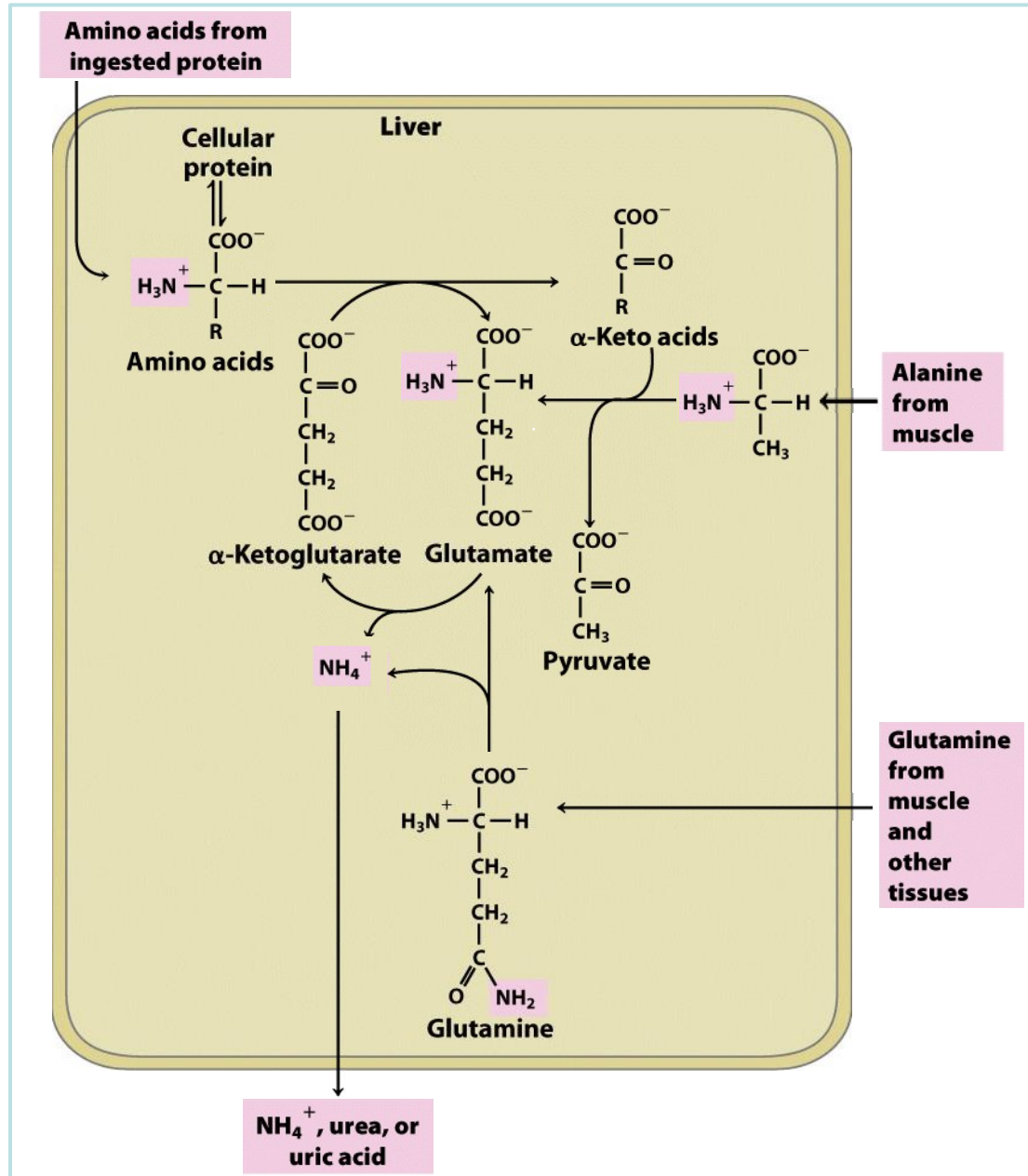
**Zymogens →
active proteases**

Enteropeptidase: activates
trypsinogen to trypsin

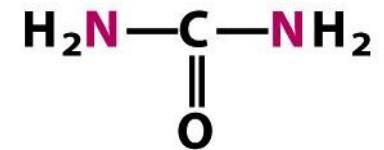
Trypsin: produce
trypsin, chymotrypsin,
carboxypeptidase,
elastase



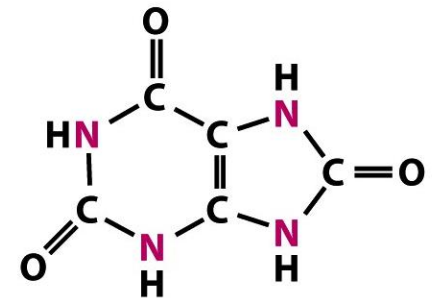
Amino group catabolism



Ammonia (as ammonium ion)
ammonotelic
(aquatic vertebrates)



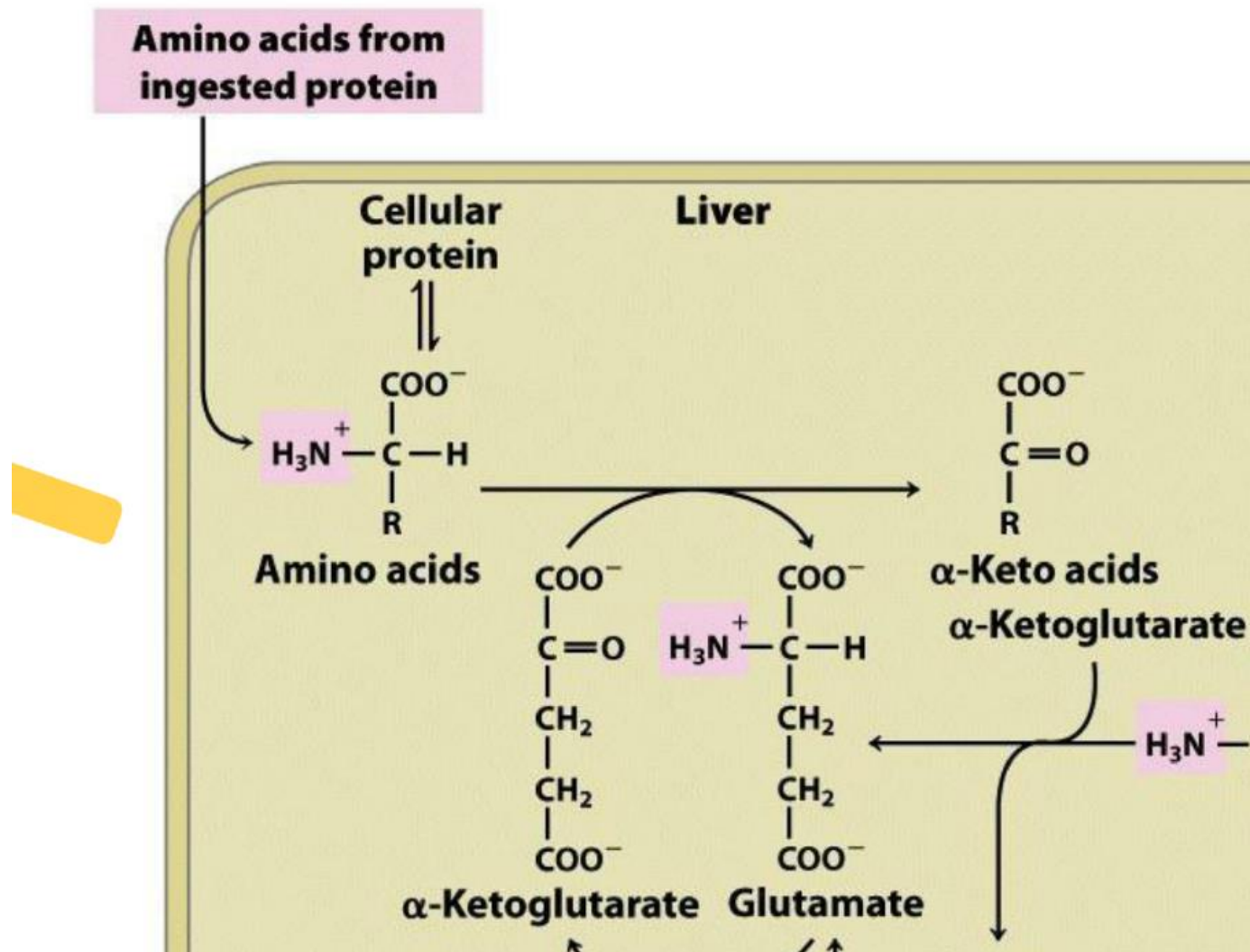
Urea
ureotelic
(terrestrial vertebrates)



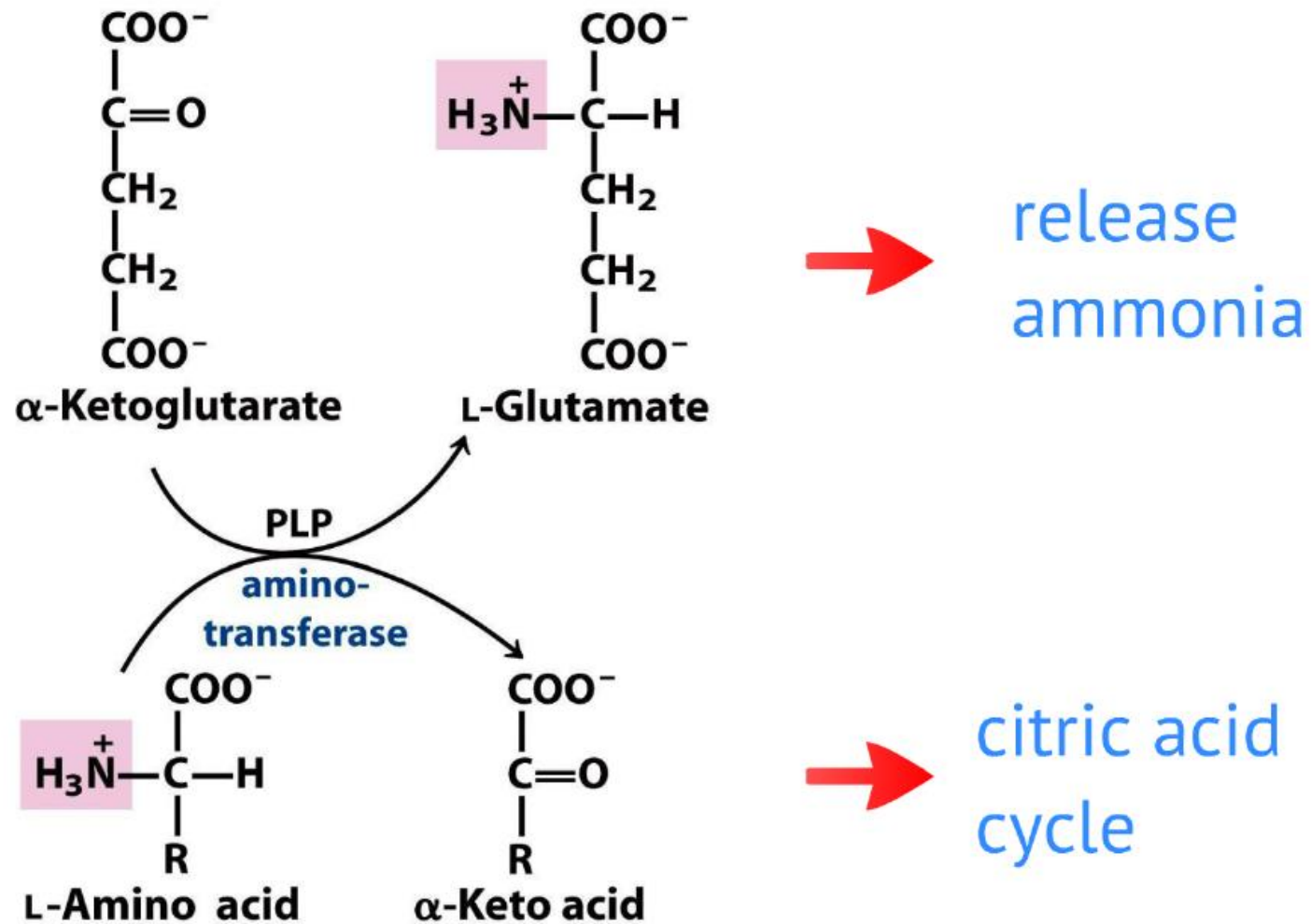
Uric acid

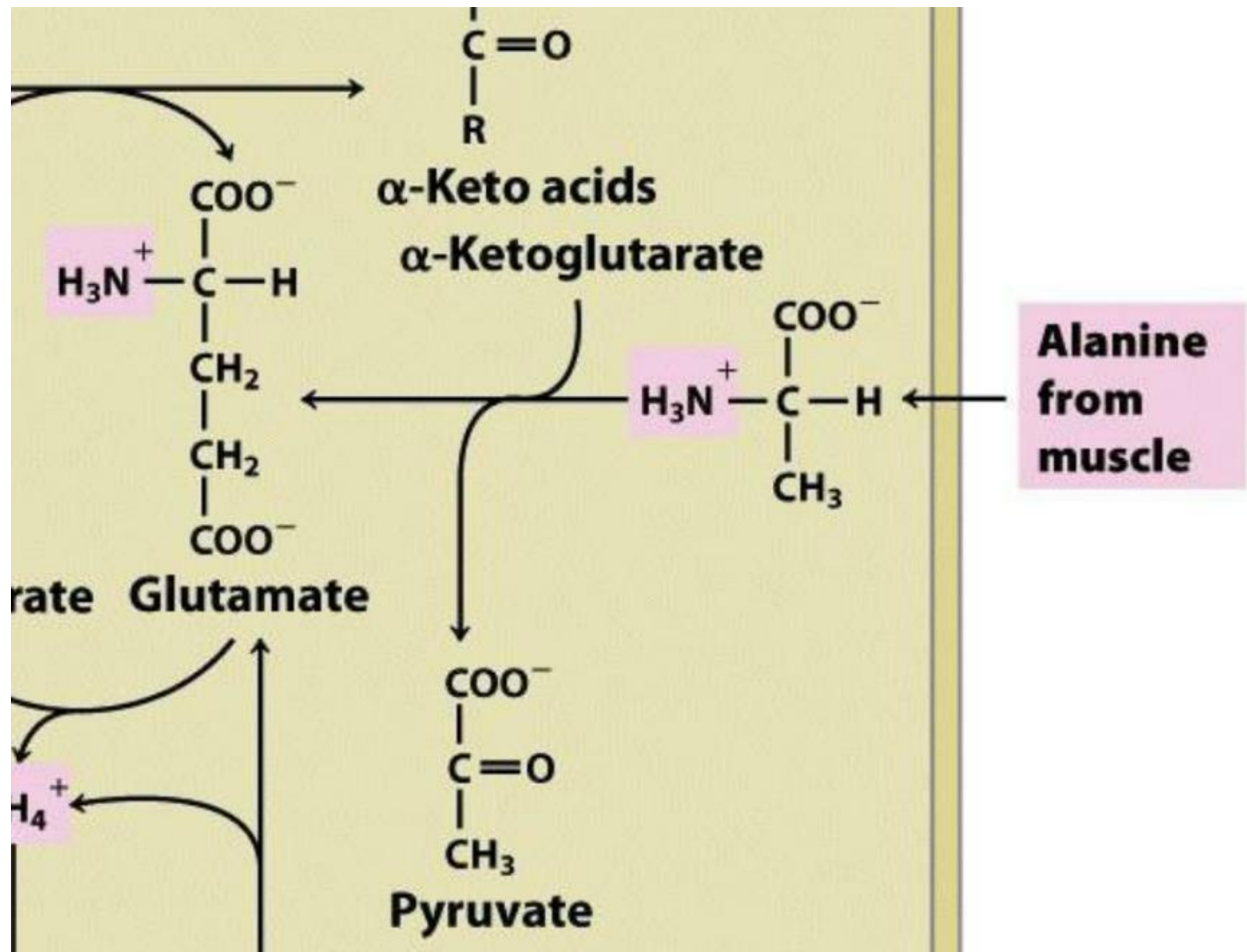
uricotelic
(birds & reptiles)

How are the uptaken amino acids being used ?

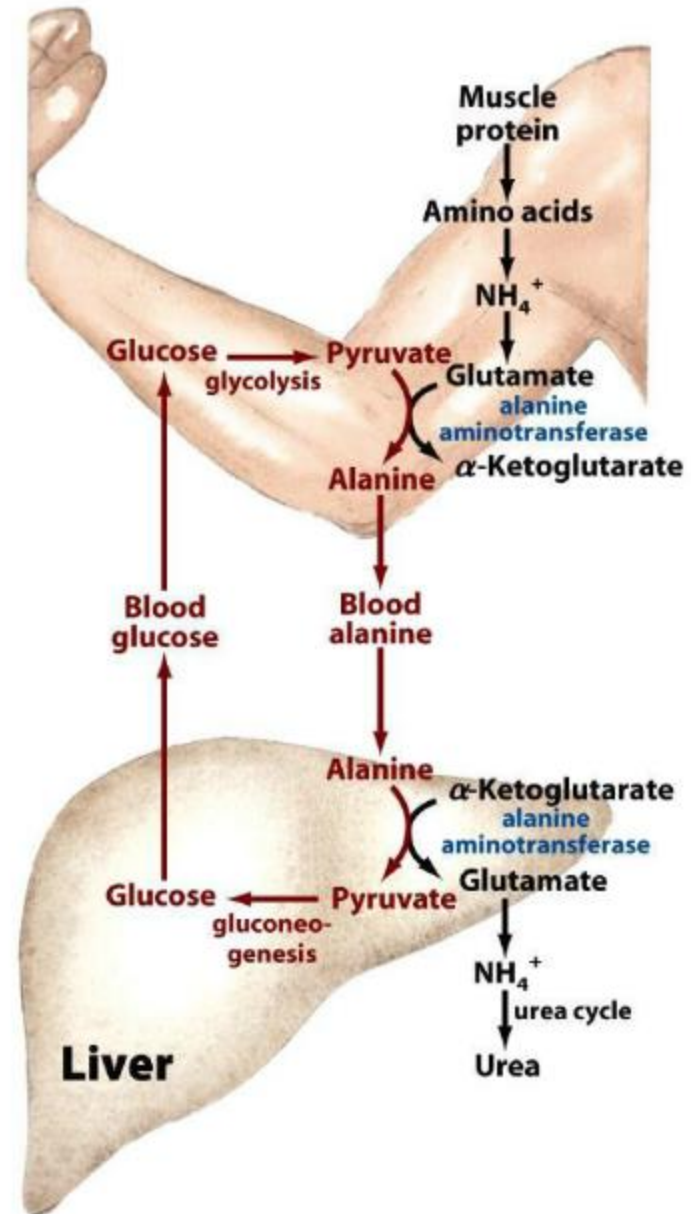


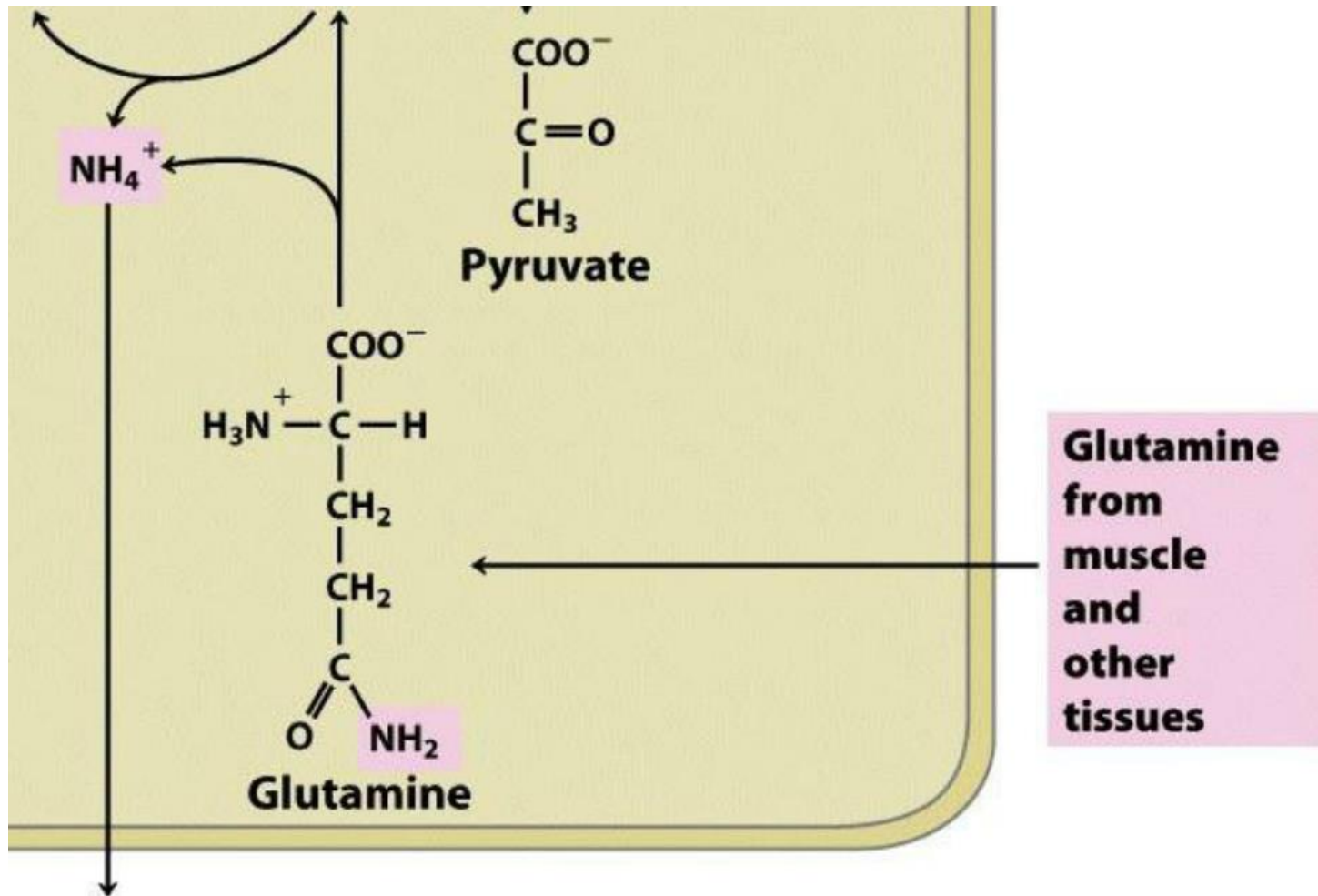
Transamination reaction by aminotransferases





alanine transport
ammonia from
skeletal muscle
to the liver
(glucose-alanine cycle)





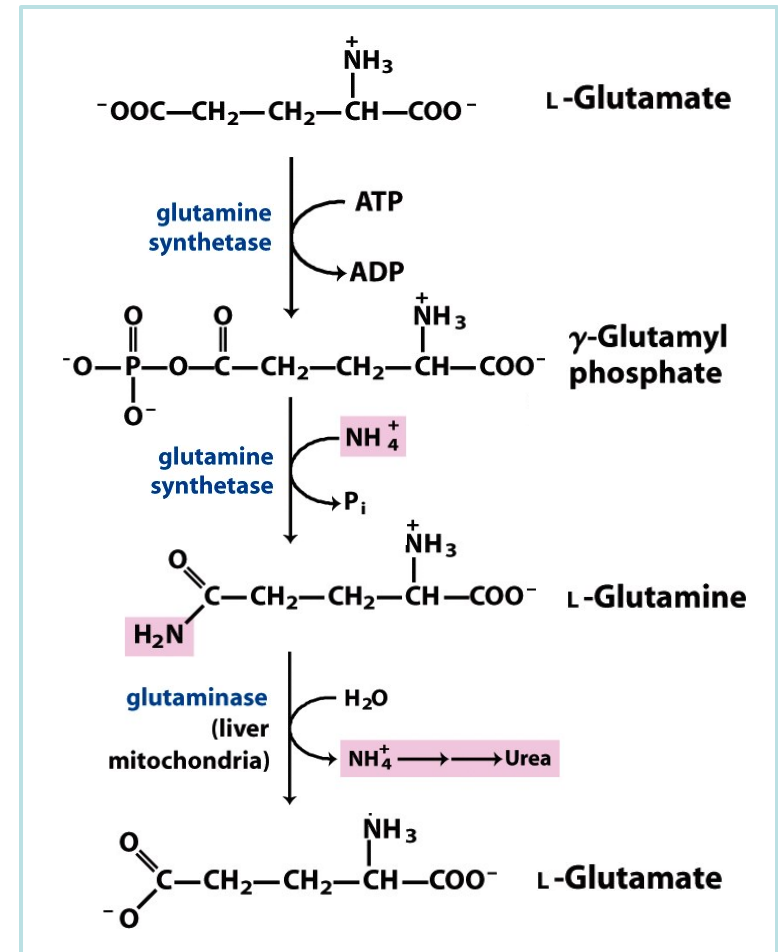
Glutamine transports ammonia in the blood stream

- Ammonia
: toxic in extrahepatic tissues
- **Gln synthetase** takes ammonia into glutamine
: safe form of ammonia

↑ [Gln] in blood;

(1) biosynthetic pathways

(2) produce NH_4^+ for Urea cycle



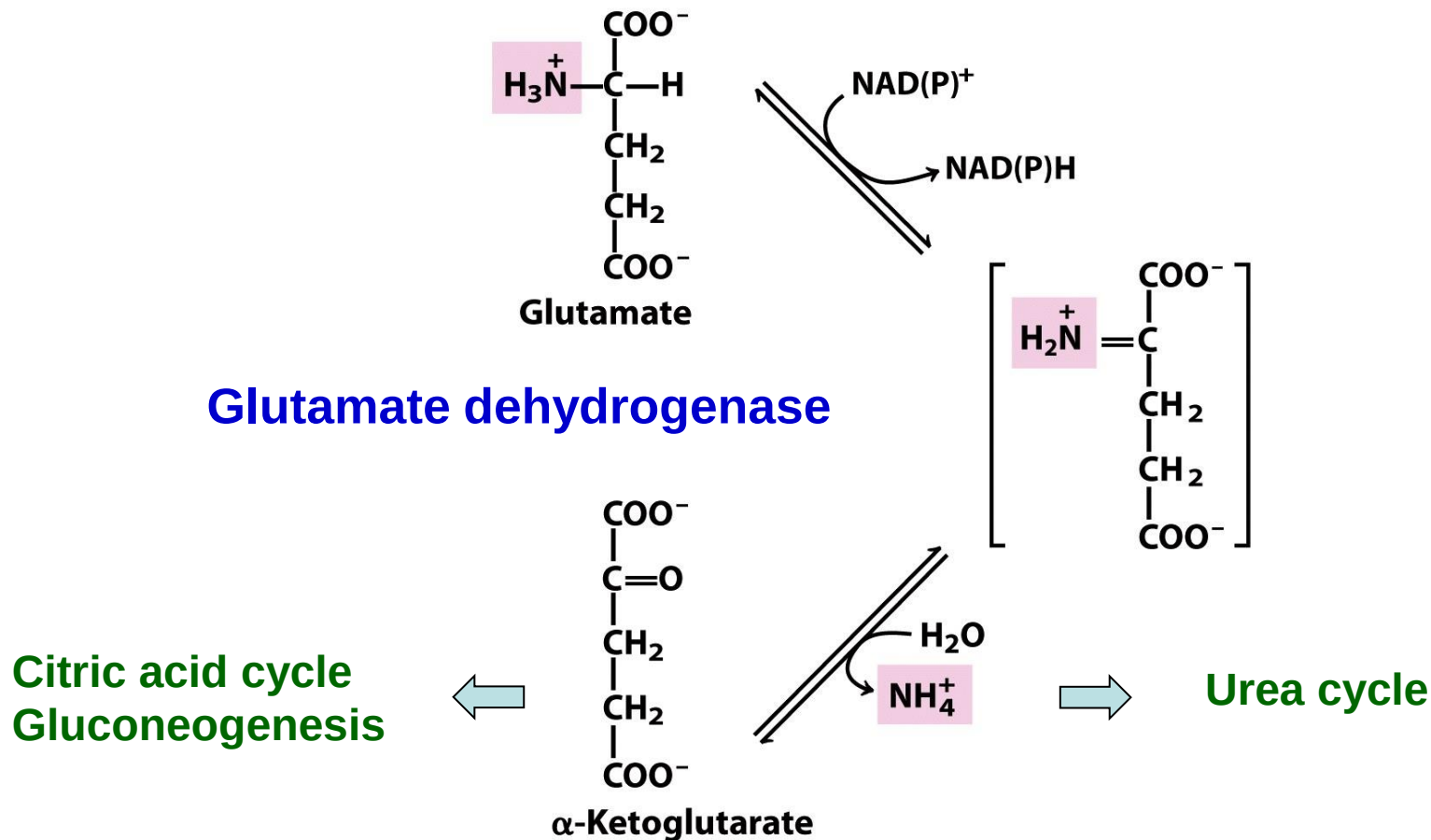
Ammonia is toxic to animals

Increased NH_4^+ $\rightarrow$ increased glutamine & reduced glutamate

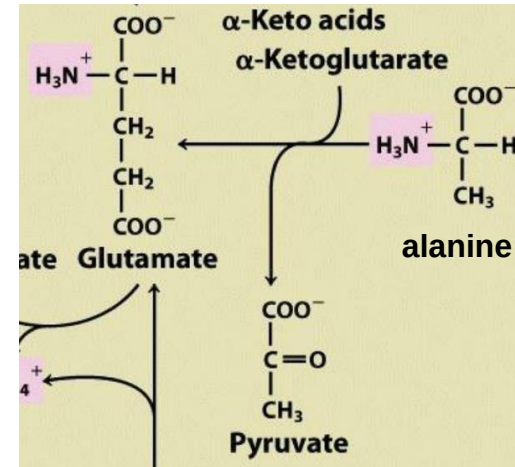
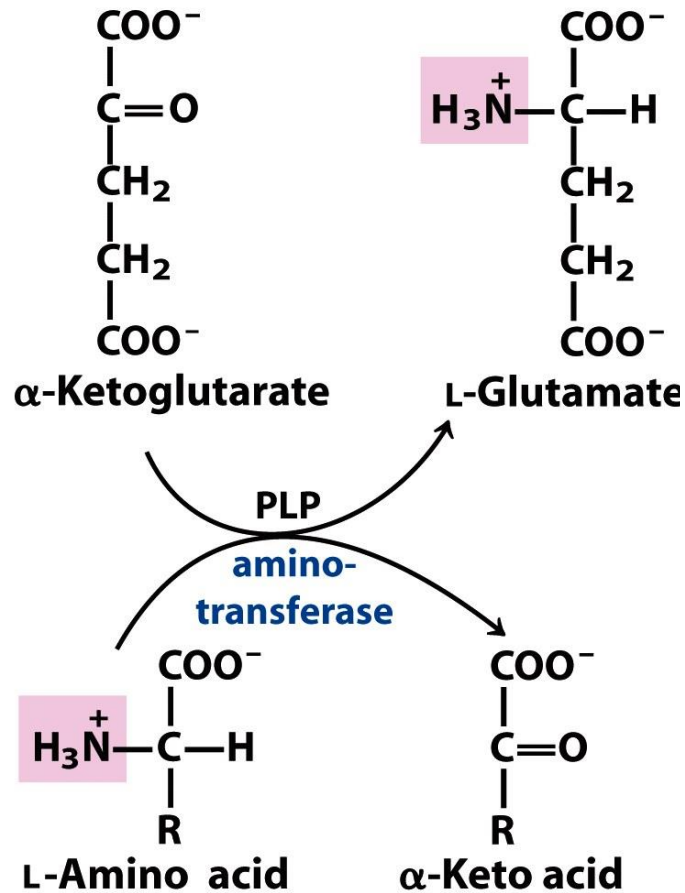
- (1) Osmotic pressure $\rightarrow$ uptake of water $\rightarrow$ brain edema $\rightarrow$ coma
- (2) Depletion of glutamate $\rightarrow$ depletion of neurotransmitter (glutamate & γ -aminobutyrate (GABA))

Glutamate release its amino group to secrete

- Oxidative deamination: in hepatocyte mitochondria
; Ammonia from glutamate ($\blacktriangle$ ADP, $\times$ GTP)

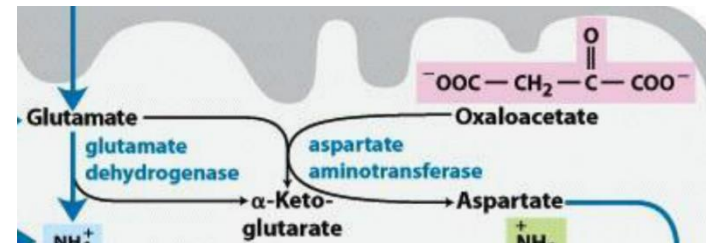


Key enzymes of AA metabolism

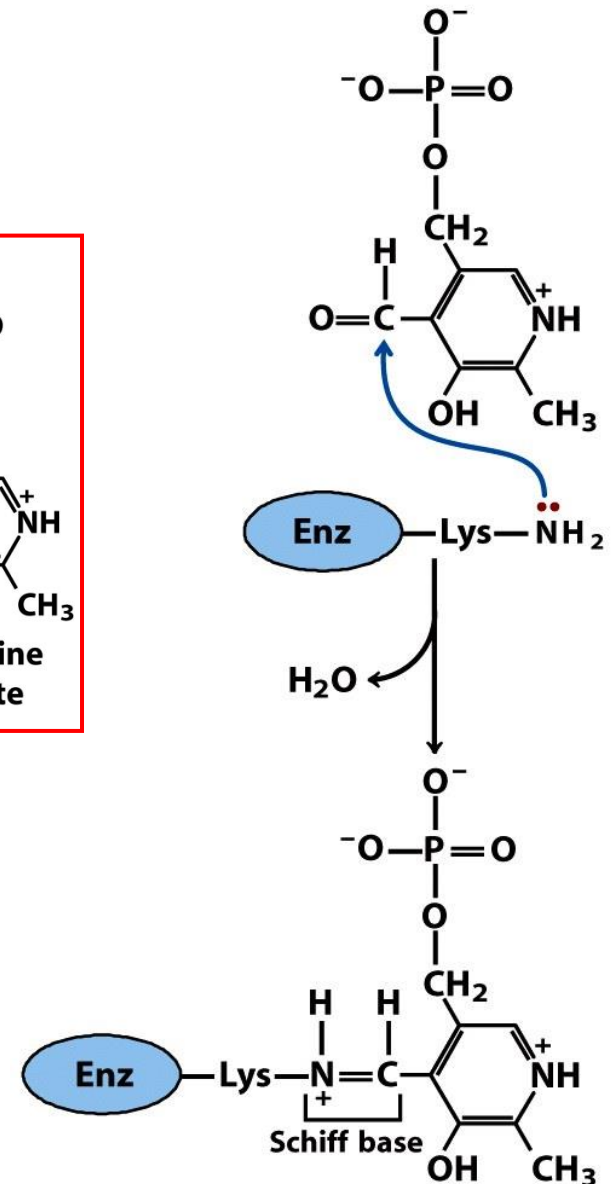
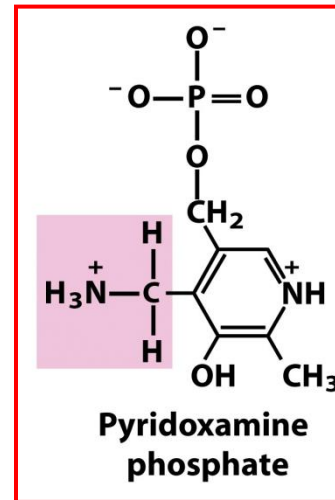
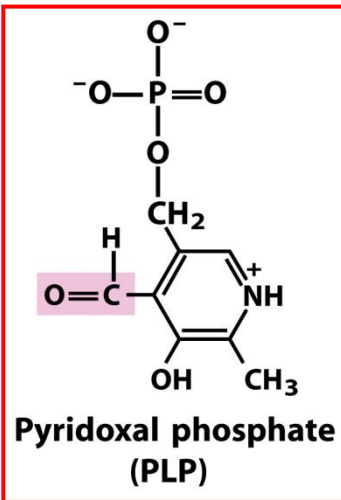
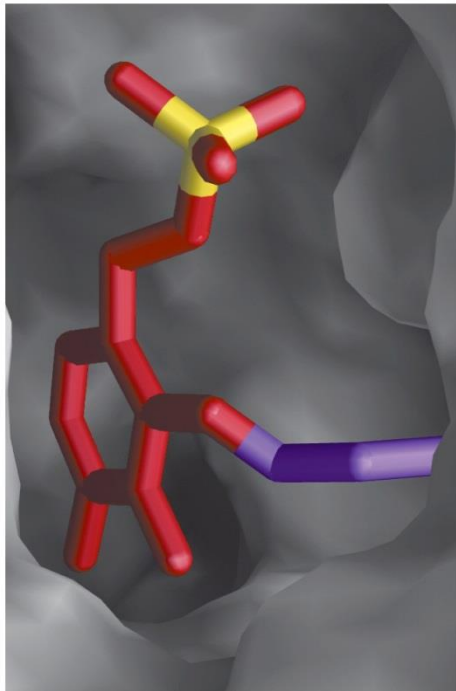


alanine aminotransferase (ALT)
(= glutamate-pyruvate transaminase, GPT)

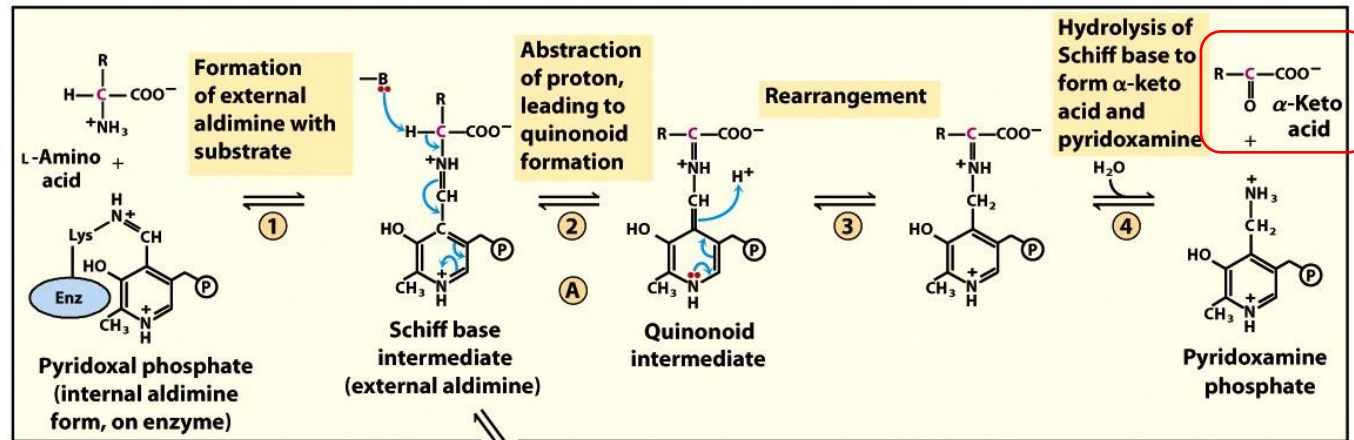
aspartate aminotransferase (AST)
(= glutamate-oxaloacetate transaminase, GOT)



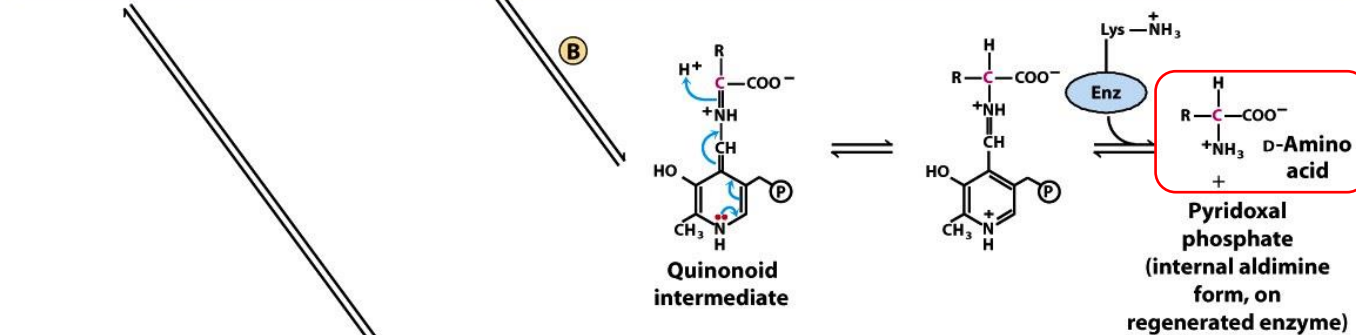
Pyridoxal phosphate (PLP), the prosthetic group of aminotransferases (transamination)



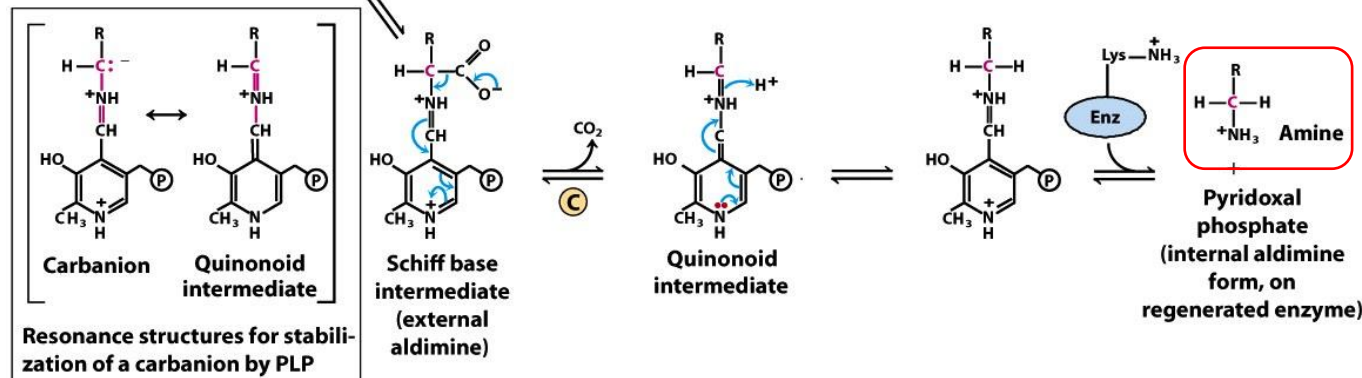
Some amino acid transformations at the α carbon that are facilitated by pyridoxal phosphate



transamination



racemization



decarboxylation

Summary

1. Dietary protein is enzymatically degraded to amino acids
2. Pyridoxal phosphate participates in the transfer of α -amino groups to α -ketoglutarate
3. Glutamate releases its amino group as ammonia in the liver
4. Glutamine transports ammonia in the bloodstream
5. Alanine transports ammonia from skeletal muscles to the liver
6. Ammonia is toxic to animals

18.2 Nitrogen excretion and the urea cycle



Nitrogen Excretion &
the Urea Cycle

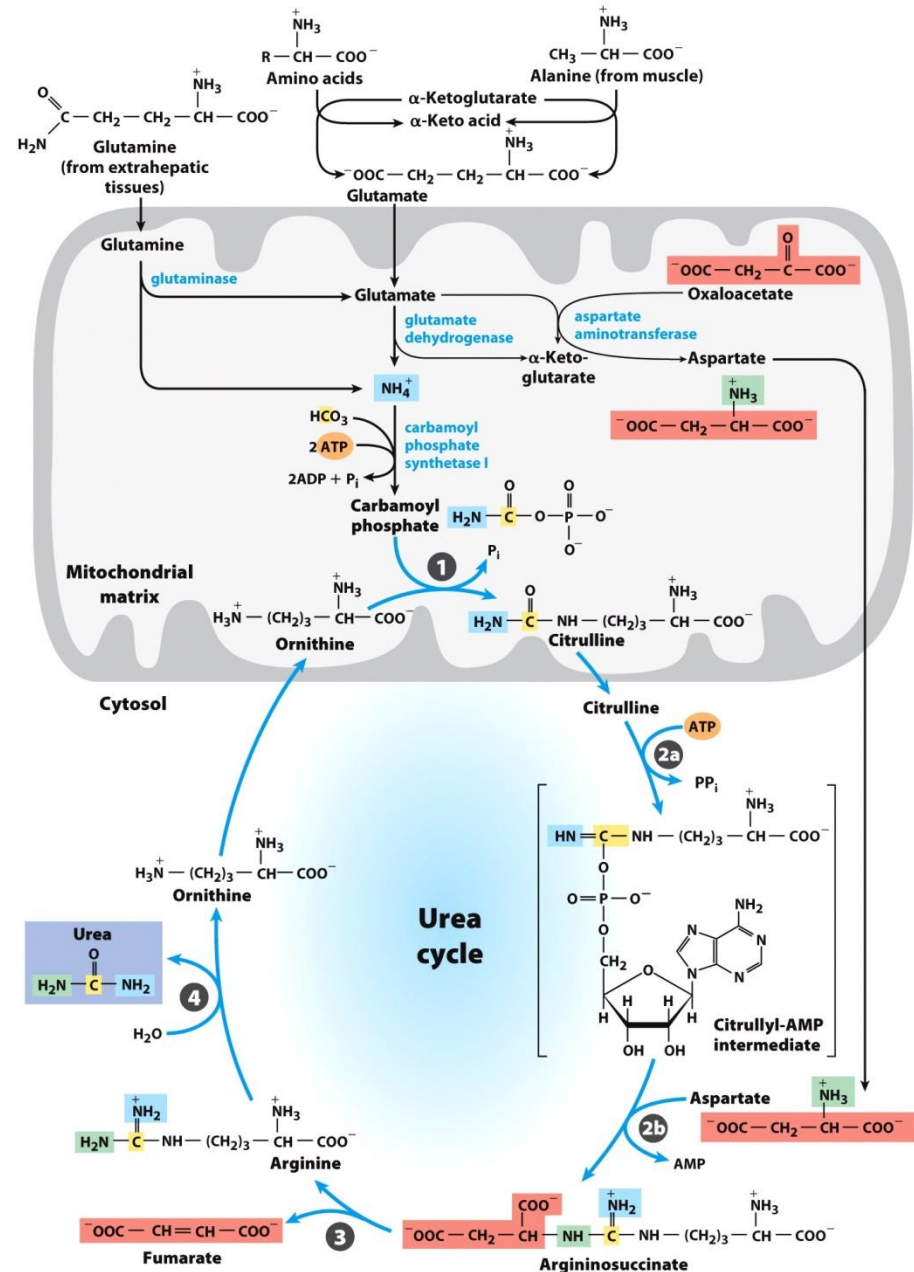
Urea is produced from ammonia by five enzymatic steps

[in mitochondria]

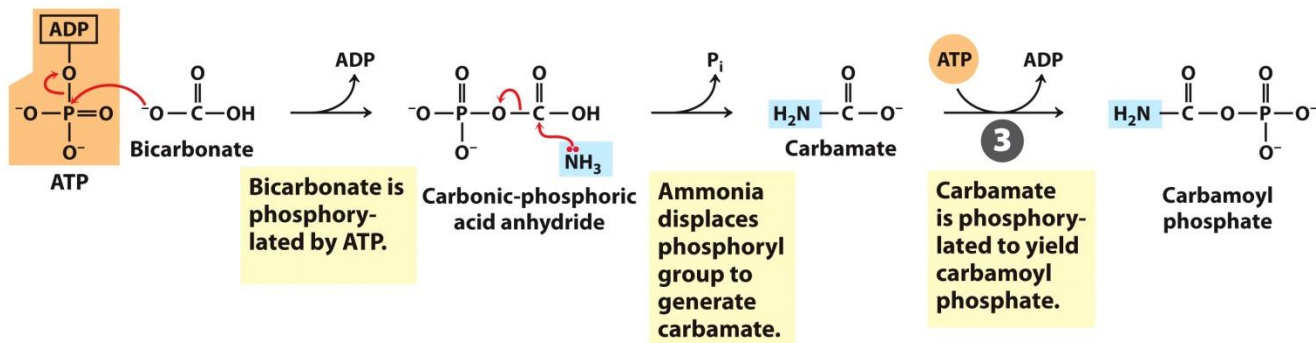
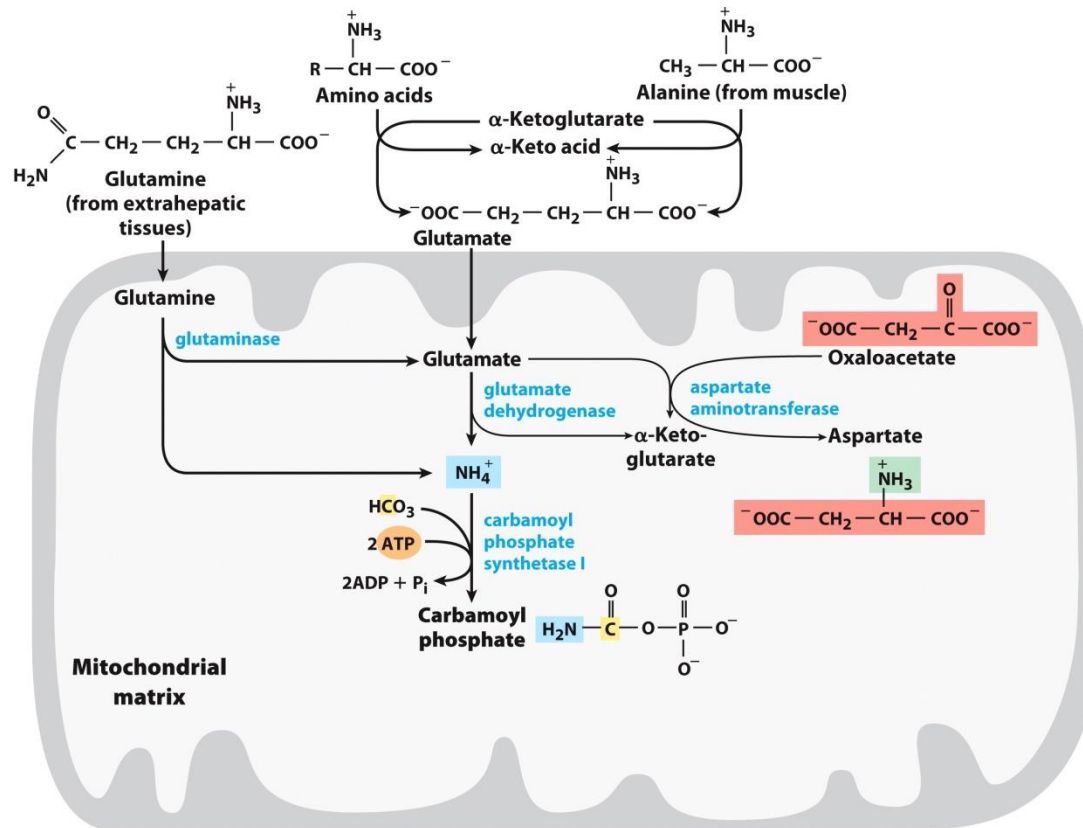
- ① Formation of **carbamoyl phosphate**: carbamoyl phosphate synthetase I
- ② Formation of **citrulline**: ornithine transcarbamoylase

[in cytosol]

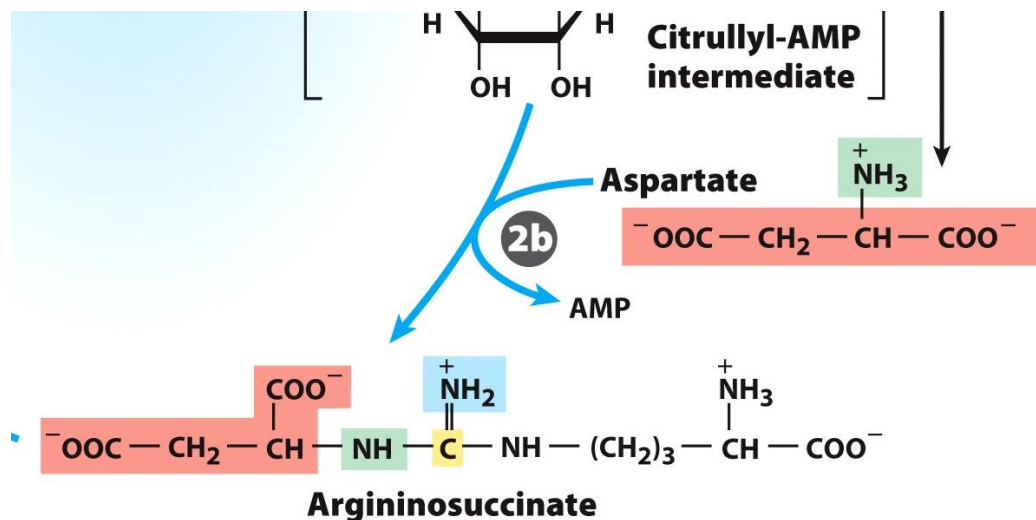
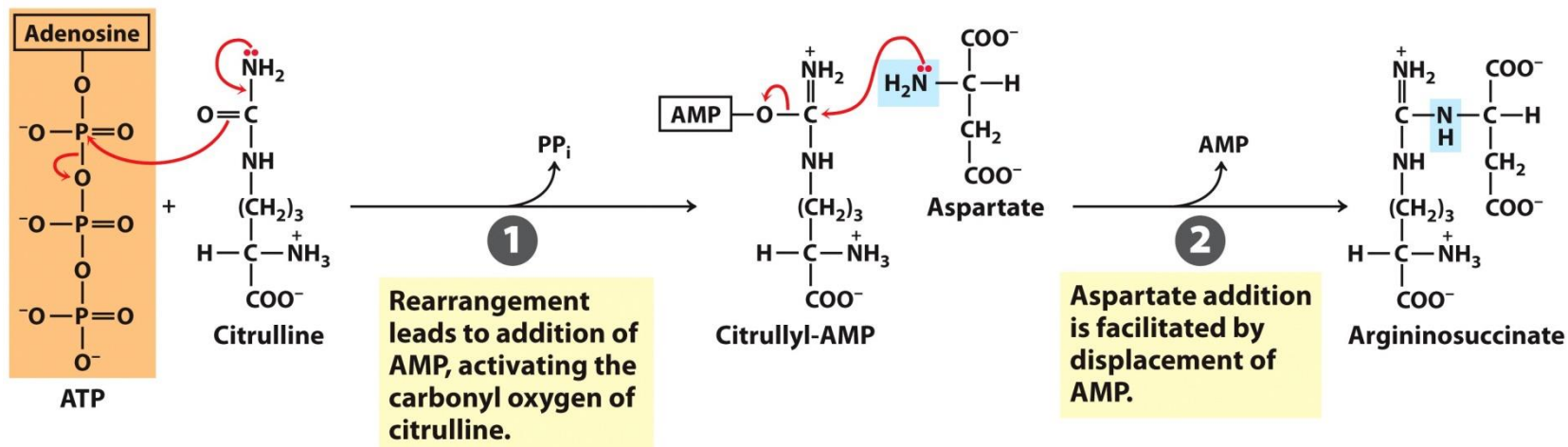
- ③ Formation of **argininosuccinate**: argininosuccinate synthetase
- ④ Formation of **arginine**: argininosuccinase
- ⑤ Formation of **urea & ornithine**: arginase



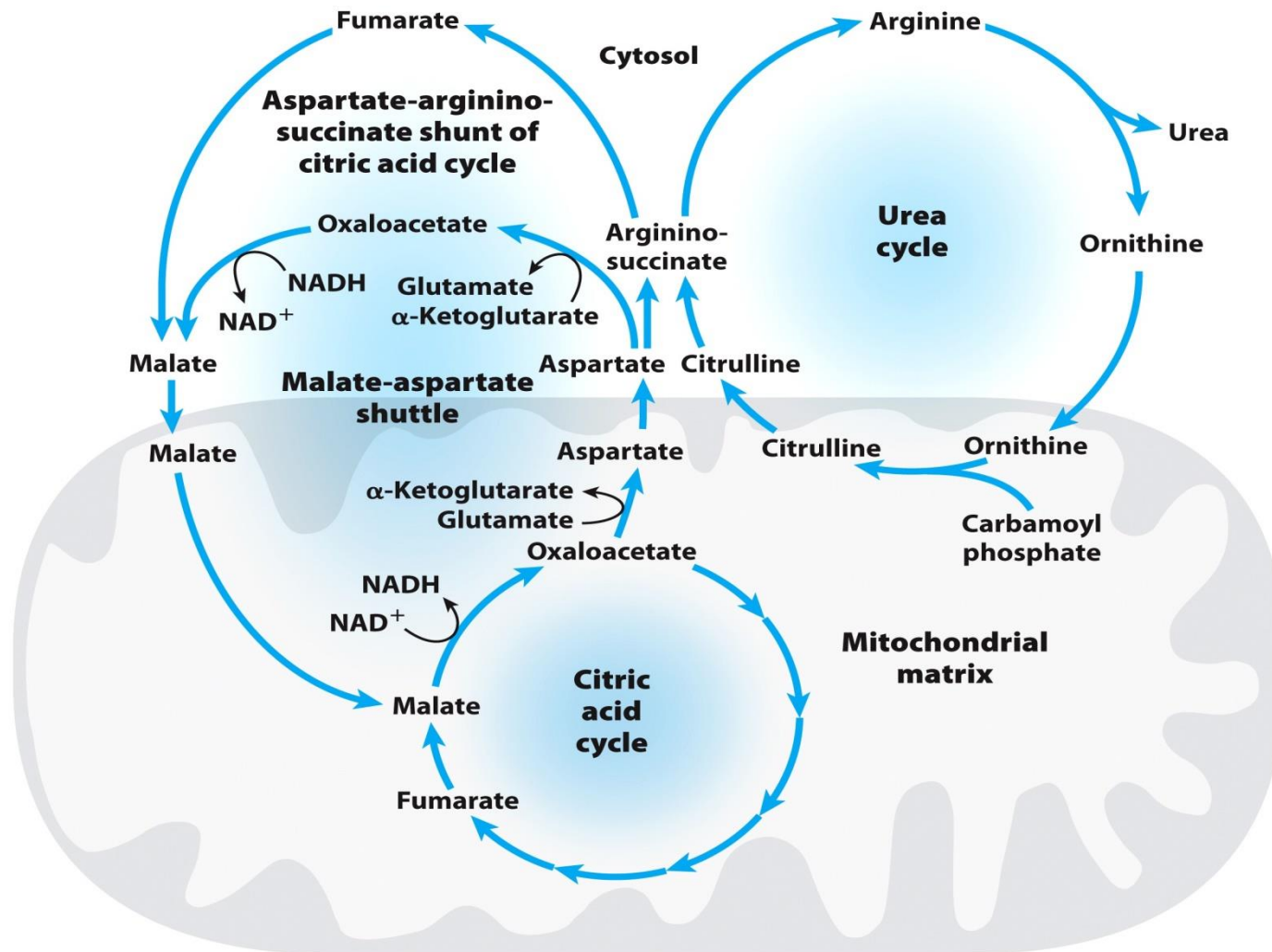
1st nitrogen enters from ammonia



2nd nitrogen enters from aspartate

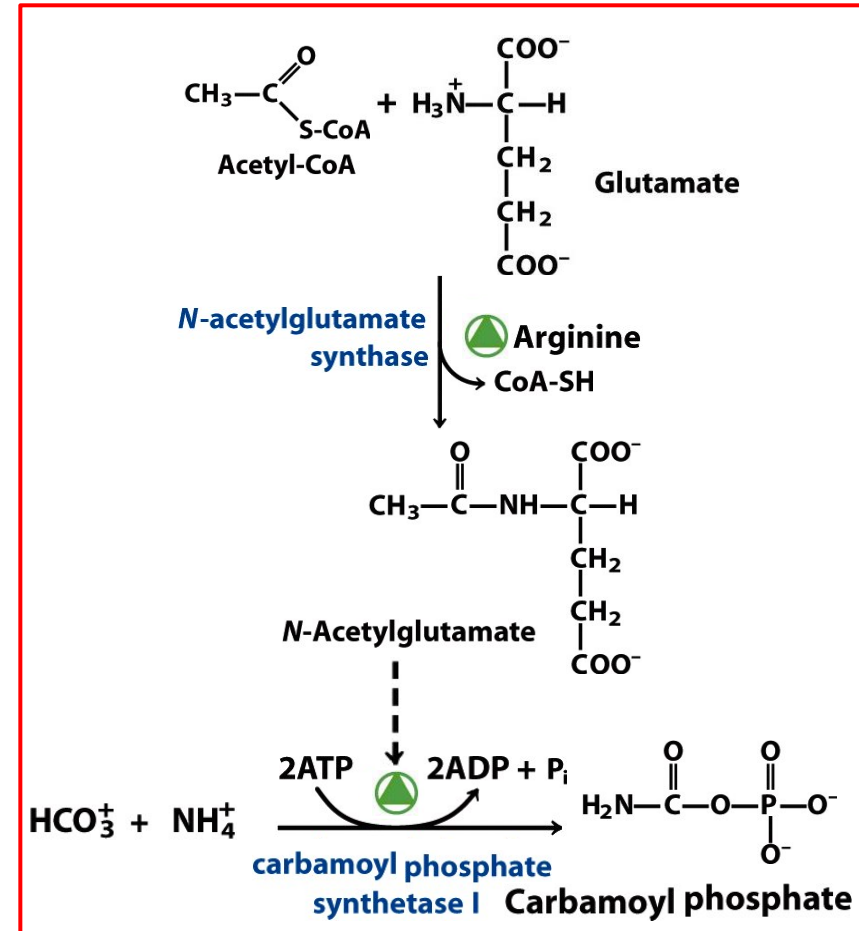


The citric acid cycle and urea cycle is linked by aspartate-argininosuccinate shunt



3) Regulation of urea cycle activity

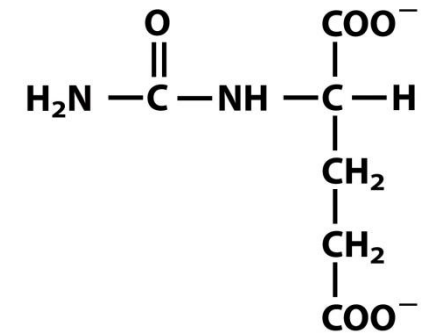
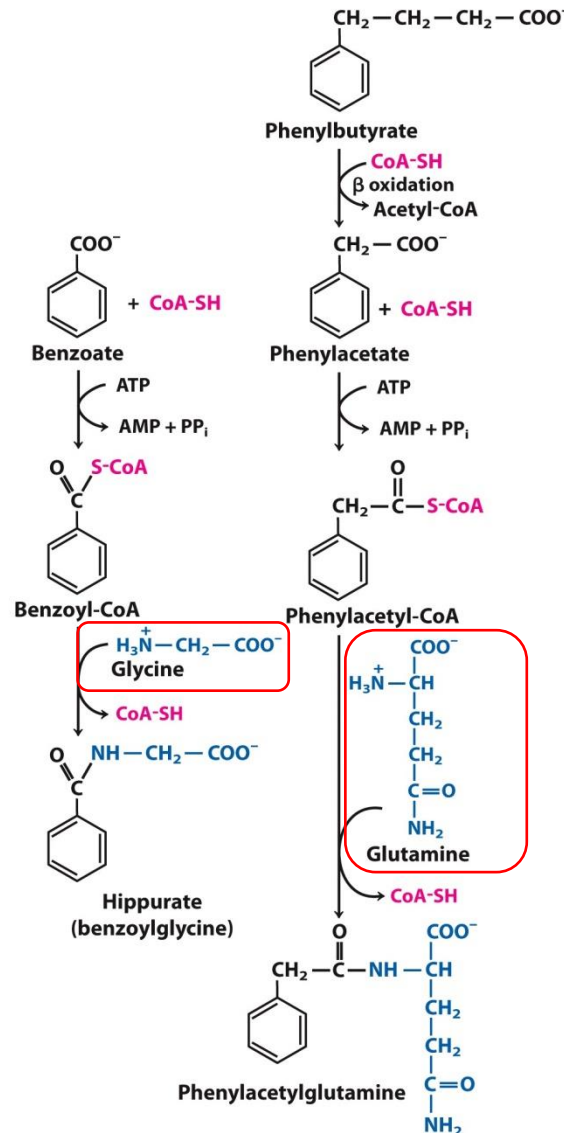
- ① All 5 enz's (carbamoyl phosphate synthetase I, ornithine transcarbamoylase, argininosuccinate synthetase, argininosuccinase, arginase): synthesis rates are increased by starvation and very-high-protein diets
- ② Carbamoyl phosphate synthetase I: activated by N-acetylglutamate



What if we have genetic defect in urea formation ?

Toxic ammonia is accumulated : hyperammonemia

Treatment of hyperammonemia by benzoate or phenylbutyrate



Carbamoyl glutamate

Activate carbamoyl
phosphate synthetase I

('22 약시)

Ammonia is removed through Gly, Gln
Products are excreted in the urine

What if we don't
take protein diet ?

: lack of essential amino
acids (HILKMFTWV)

assays for tissue damage

alanine aminotransferase (ALT)
(= glutamate-pyruvate transaminase, GPT)

aspartate aminotransferase (AST)
(= glutamate-oxaloacetate transaminase, GOT)

after **heart attack** : SCK, SGOT, SGPT
leak from injured heart cells

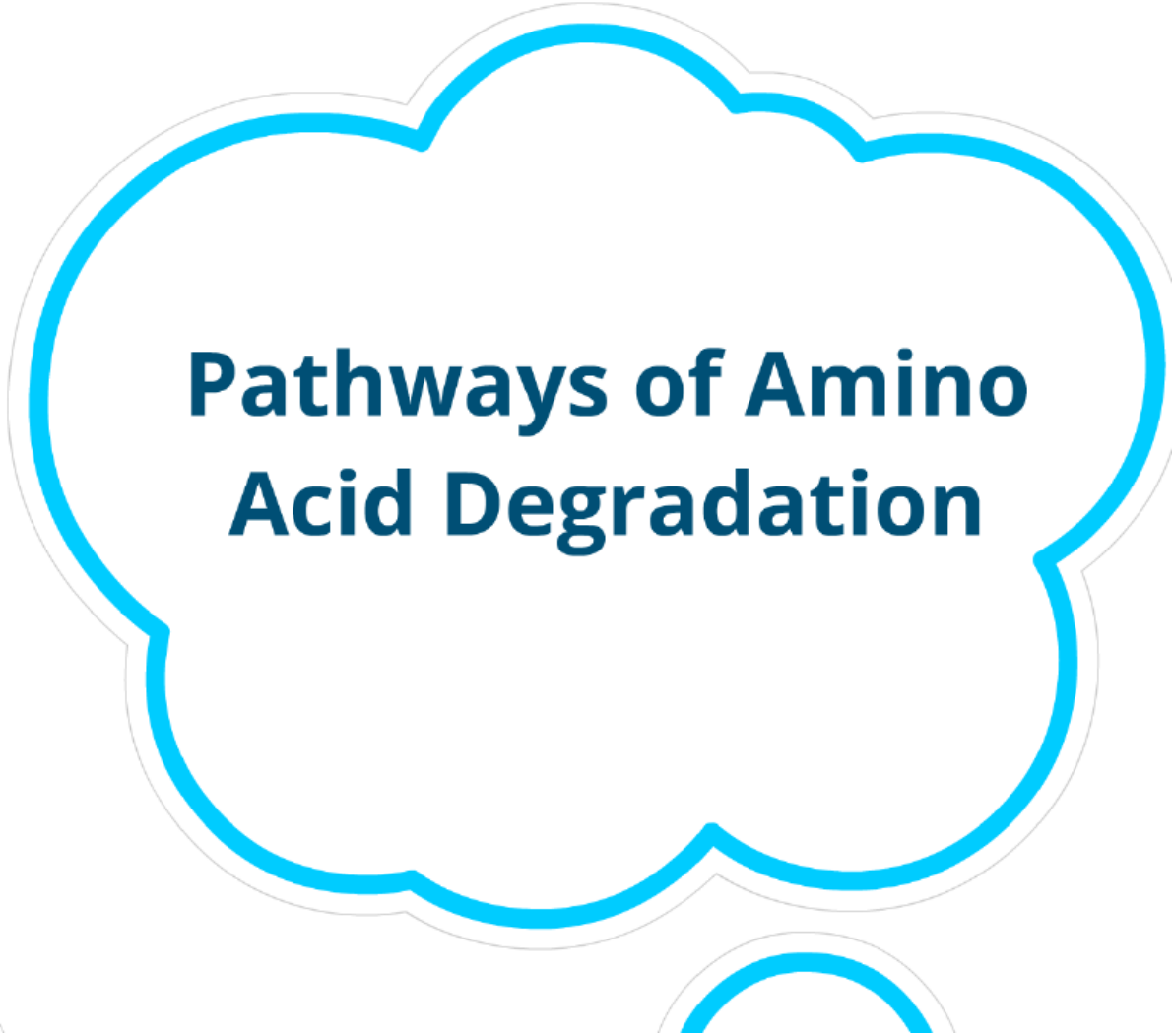
liver damage by CCl₄ or CHCl₃: SGOT,
SGPT leak from injured hepatocytes

Summary

1. Urea is produced from ammonia in five enzymatic steps
2. The citric acid and urea cycles can be linked
3. The activity of the urea cycle is regulated at two levels
4. Pathway interconnections reduce the energetic cost of urea synthesis
5. Genetic defects in the urea cycle can be life-threatening

실패할까 걱정하는 심리는 현재 자신이 하는 일에
집중하지 못하게 하고 기본적인 대응조차
심사숙고하게 (예를 들어 왼발을 먼저 내디녀야 할지,
오른발을 내디녀야 할지 고민하는 것) 만든다. 또한
이 때문에 발생하는 결과에 반응하는 속도 역시
느려져 생각을 둔하게 만든다.... (월렌다 효과)

18.3 Pathways of Amino Acid Degradation



**Pathways of Amino
Acid Degradation**

Ketogenic and glucogenic amino acids

'22 약시

LWFKY

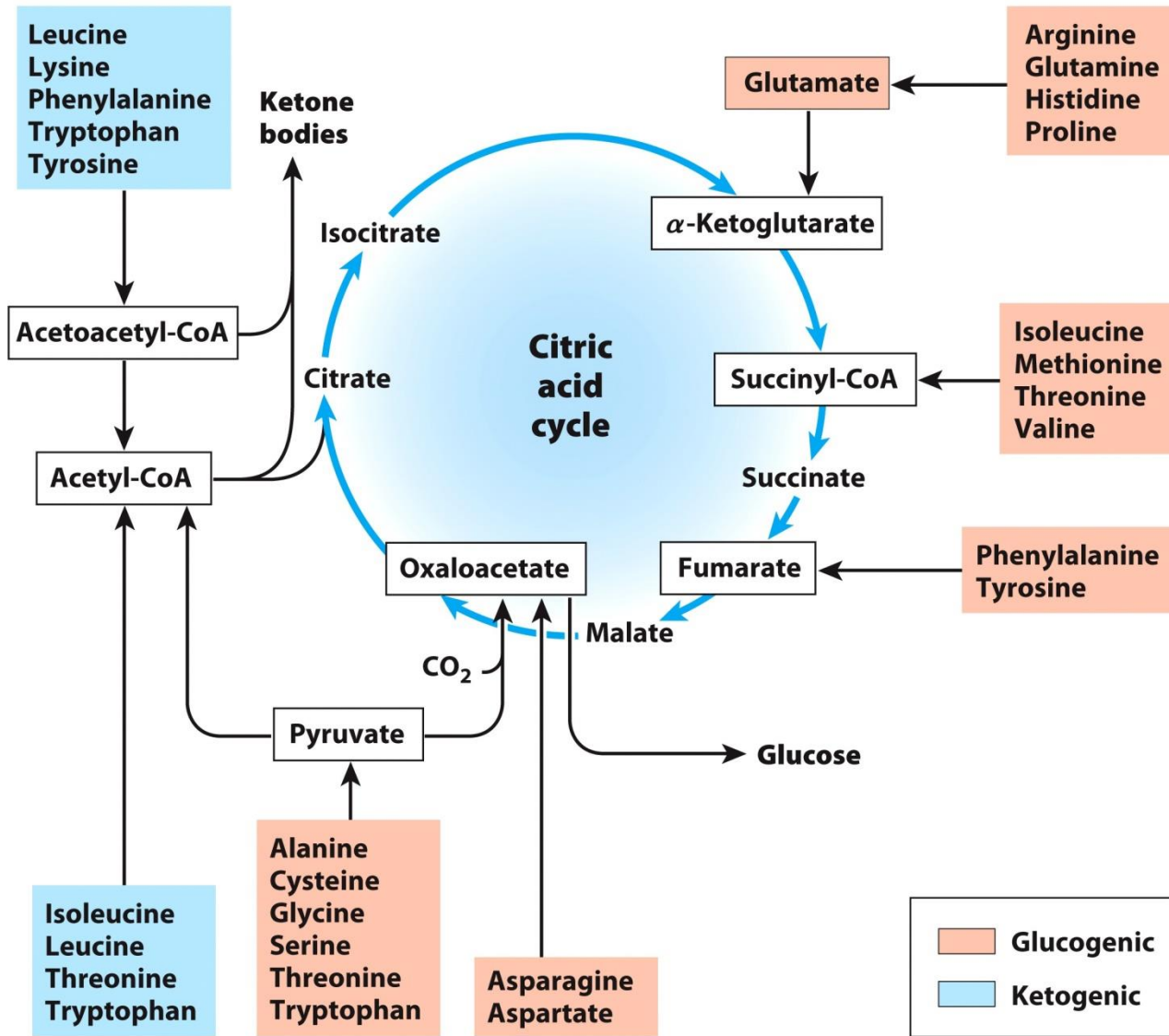
PRHEQ

IMTV

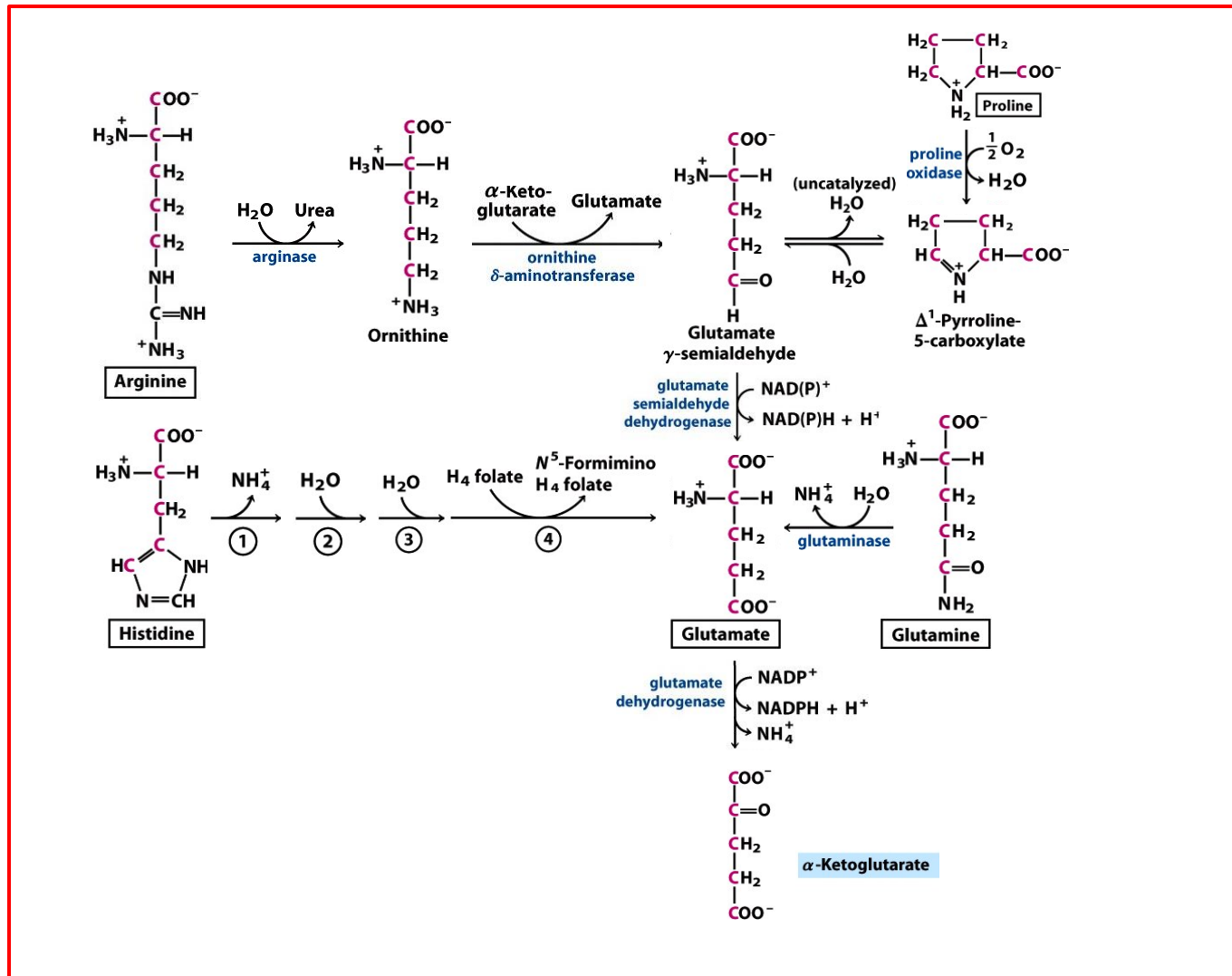
FY

LWIT

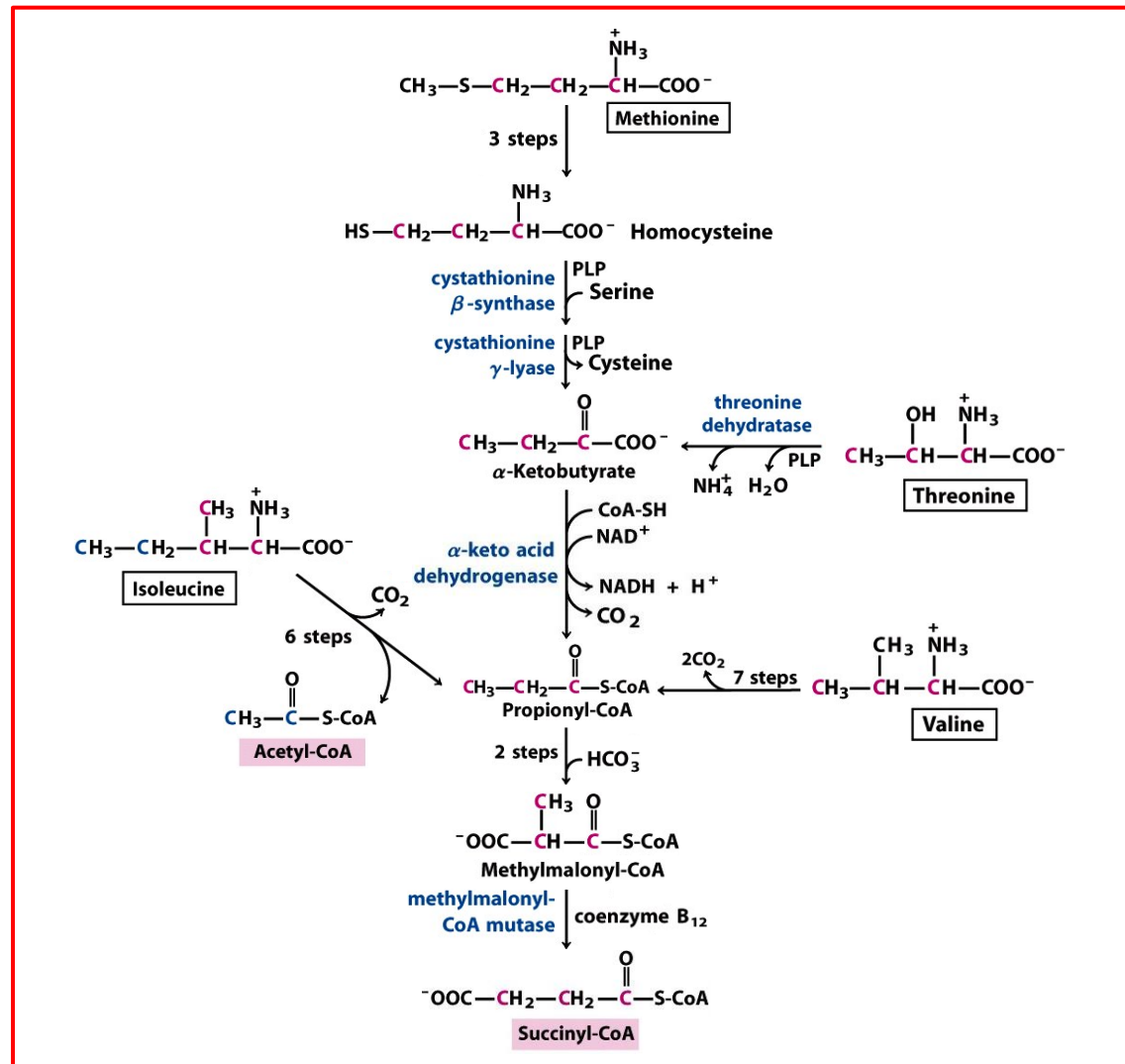
ACWGST ND



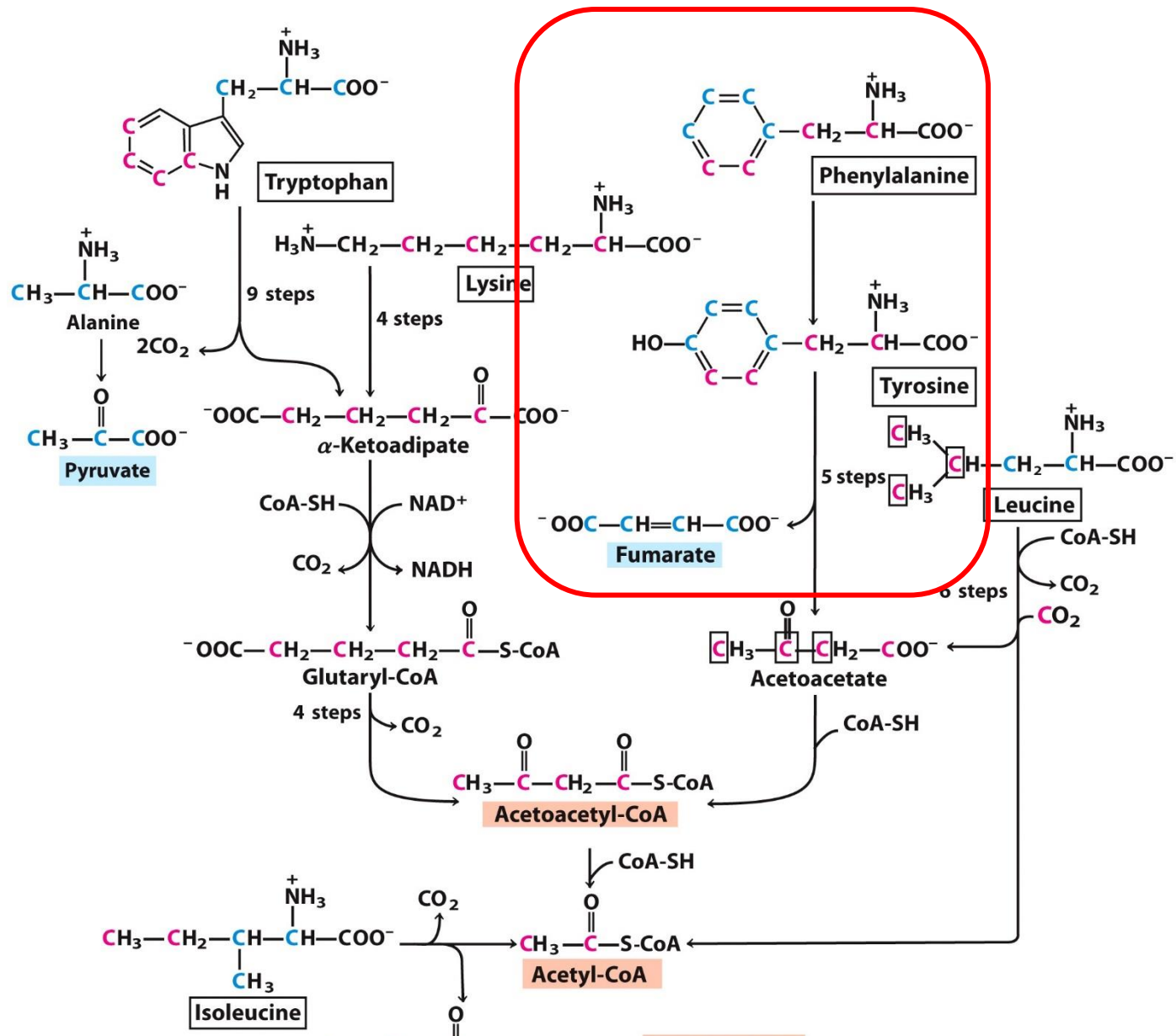
1) 5 amino acids to α -kg (PRHEQ)



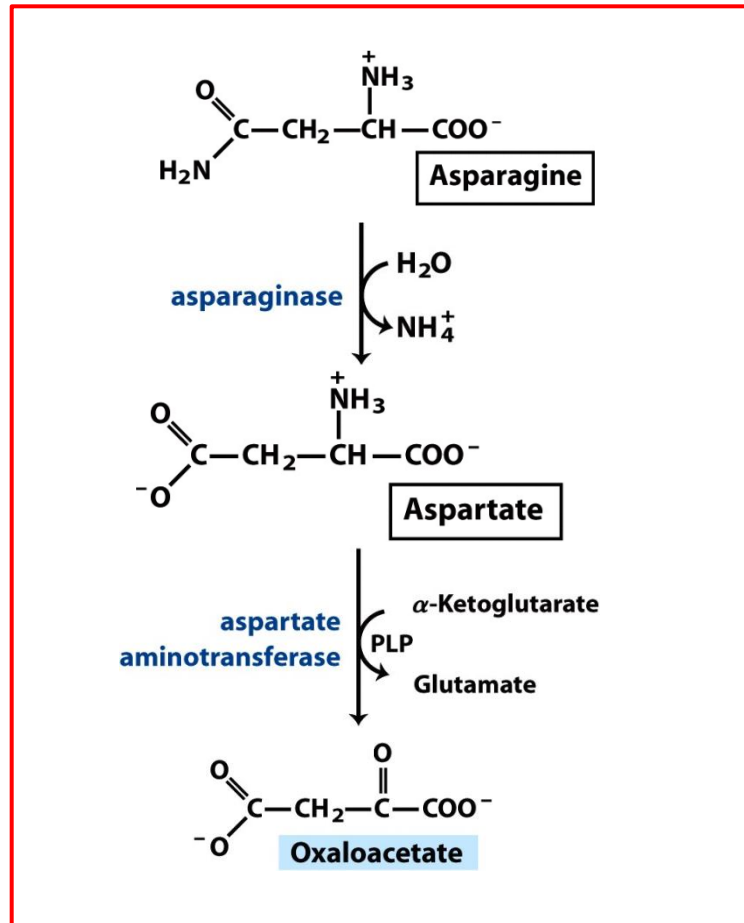
2) 4 amino acids to succinyl-CoA (IMTV)



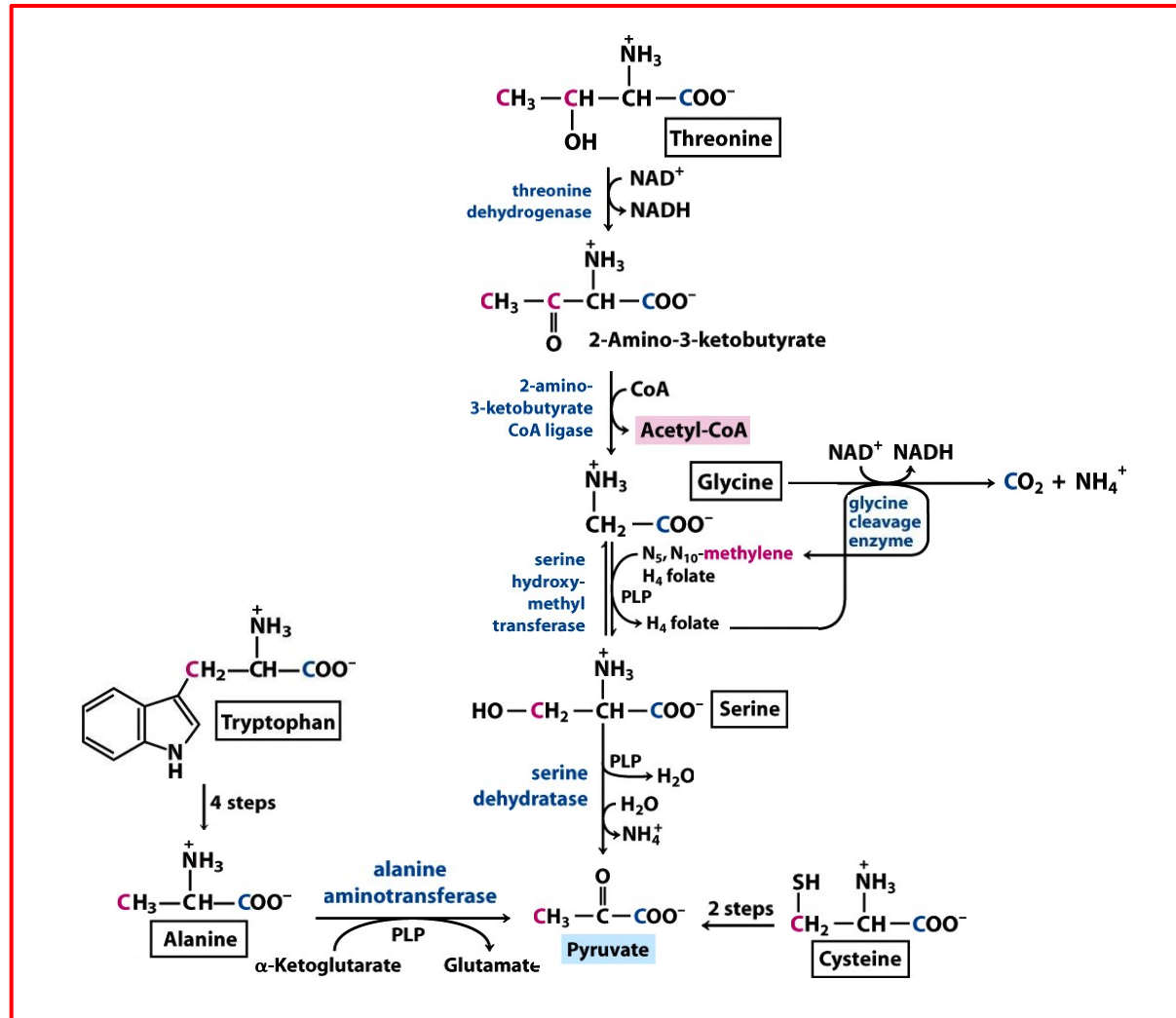
3) 2 amino acids to Fumarate (FY)



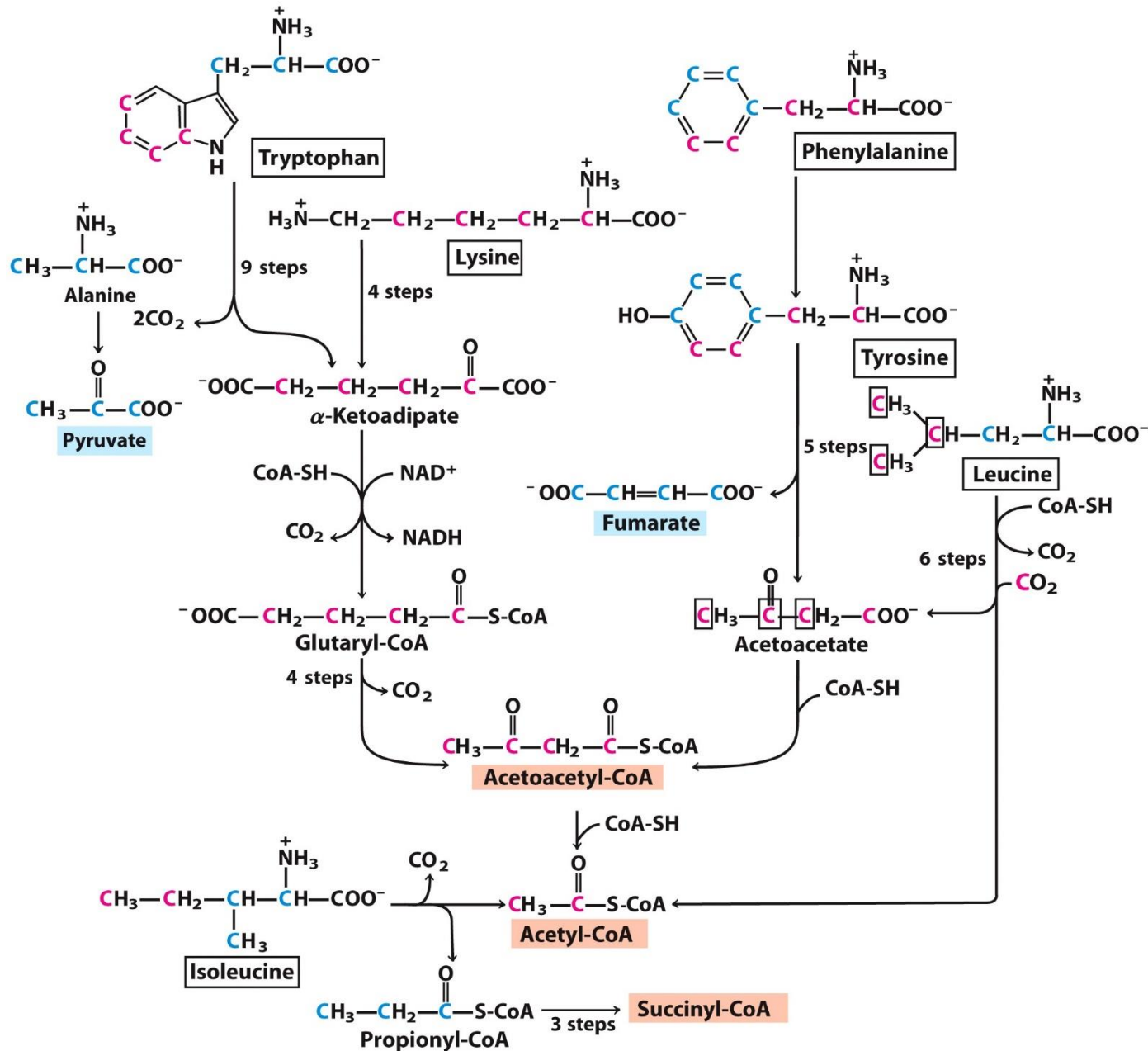
4) 2 amino acids to OAA (ND)



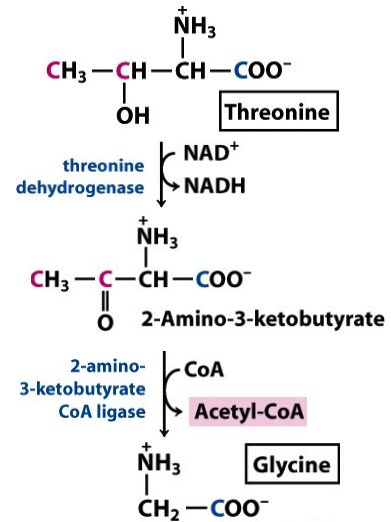
5) 6 amino acids to pyruvate (ACWGST)



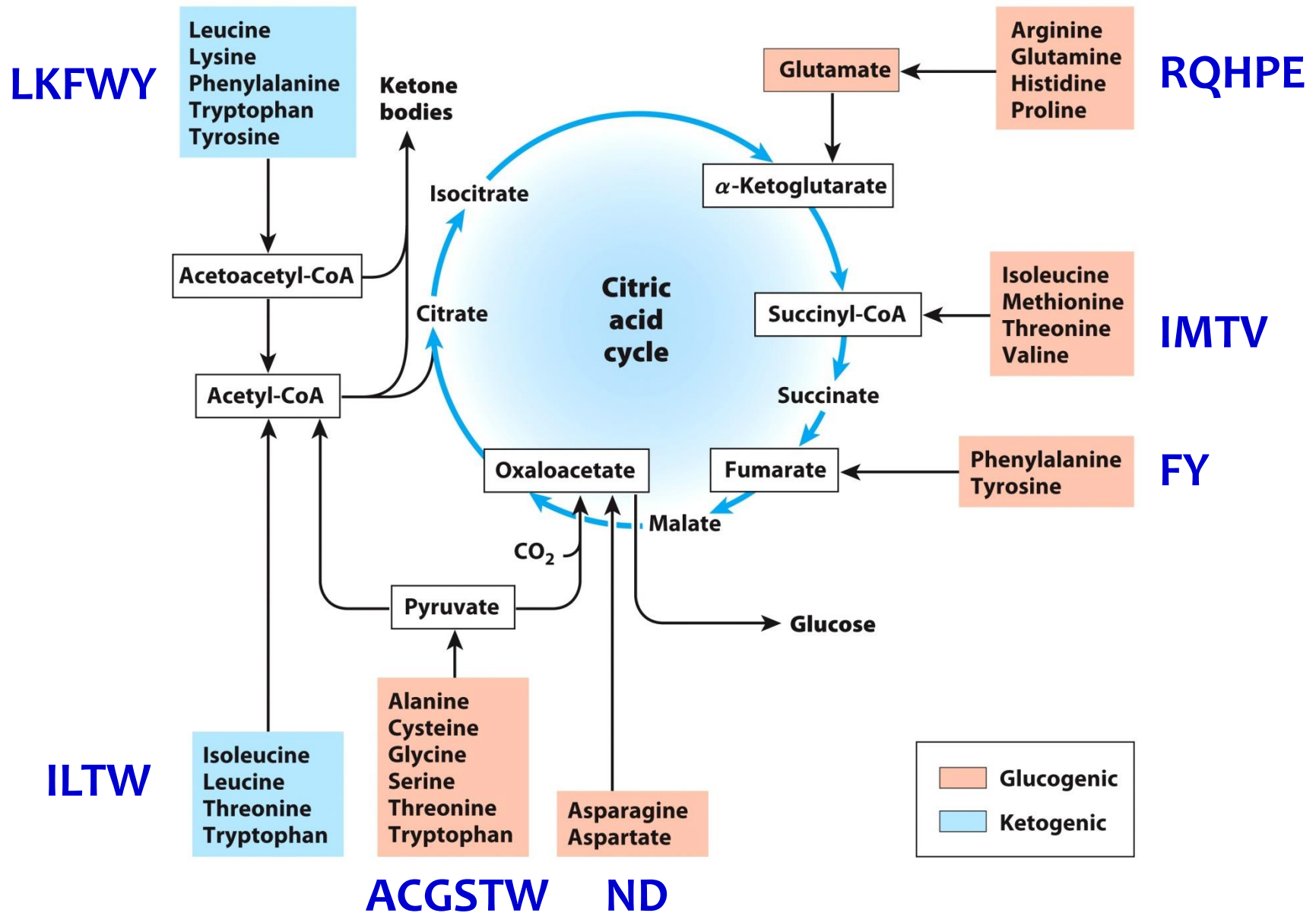
6) 6 amino acids to acetyl CoA (LWIFYKI)



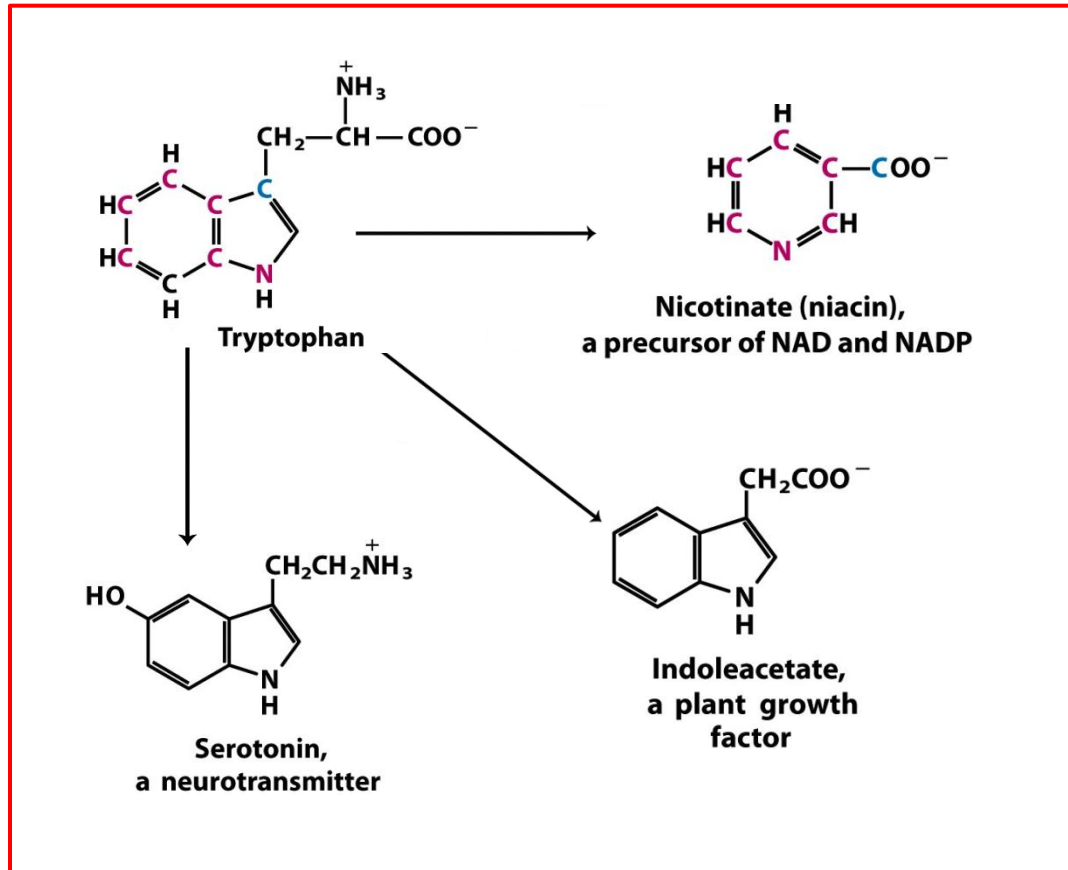
7) Threonine to acetyl CoA



Ketogenic and glucogenic amino acids

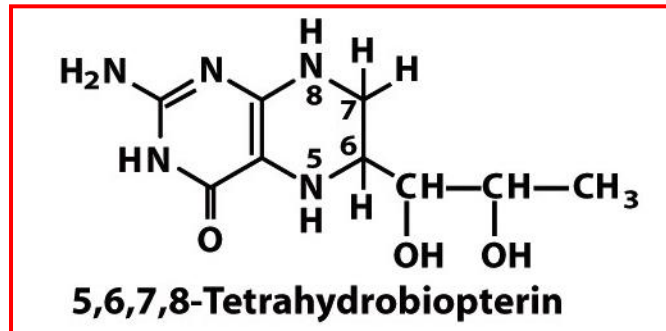


Trp as precursors



Enz cofactors in aa catabolism

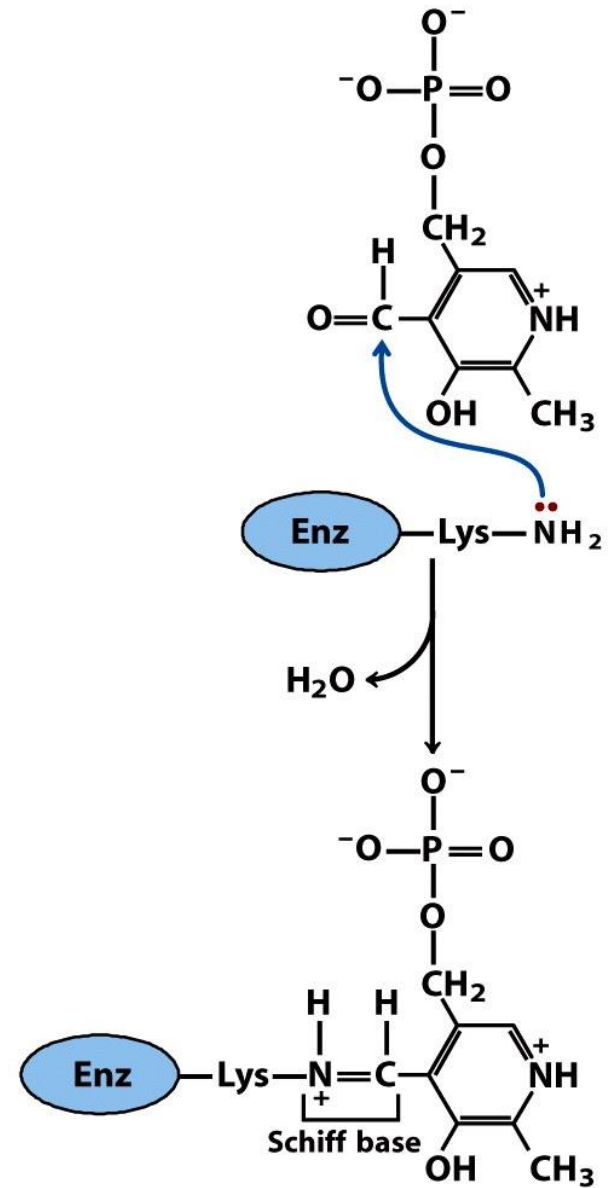
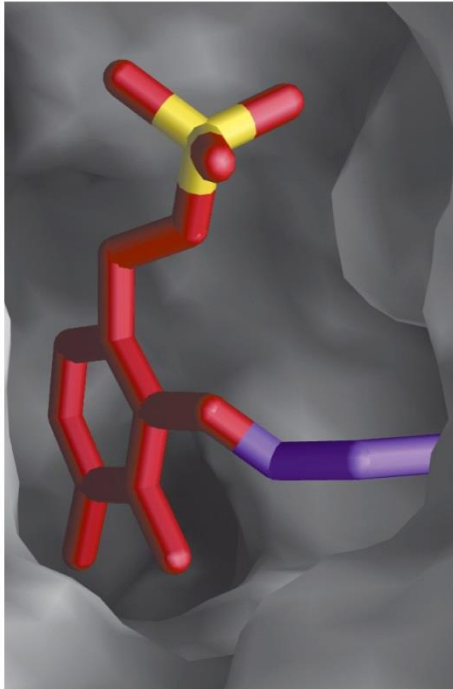
- **Transamination: Pyridoxal phosphate (PLP)**
- **1 C grp transfer: biotin (CO₂), tetrahydrofolate (CH₃/CH₂OH/CHO), S-adenosylmethionine (adoMet; CH₃)**
- **Oxidation rxn: tetrahydrobiopterin**



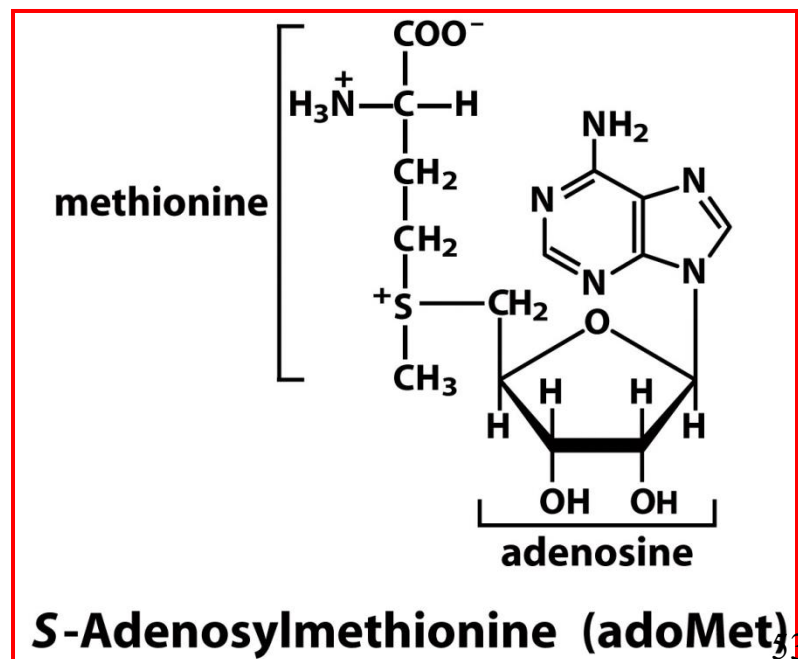
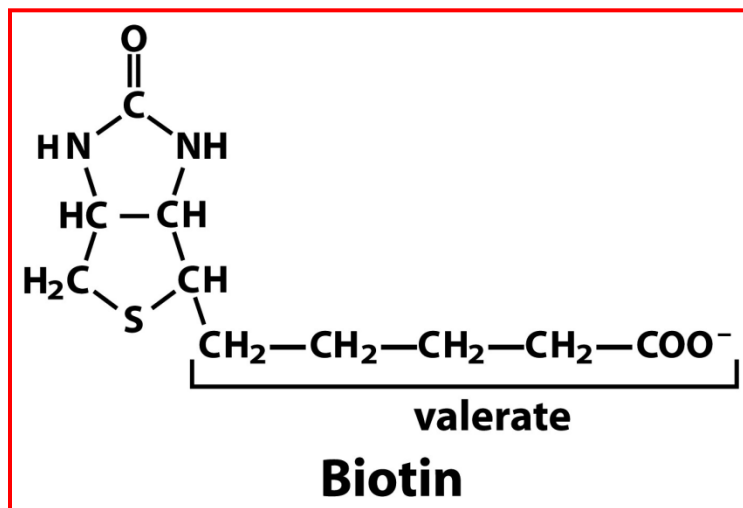
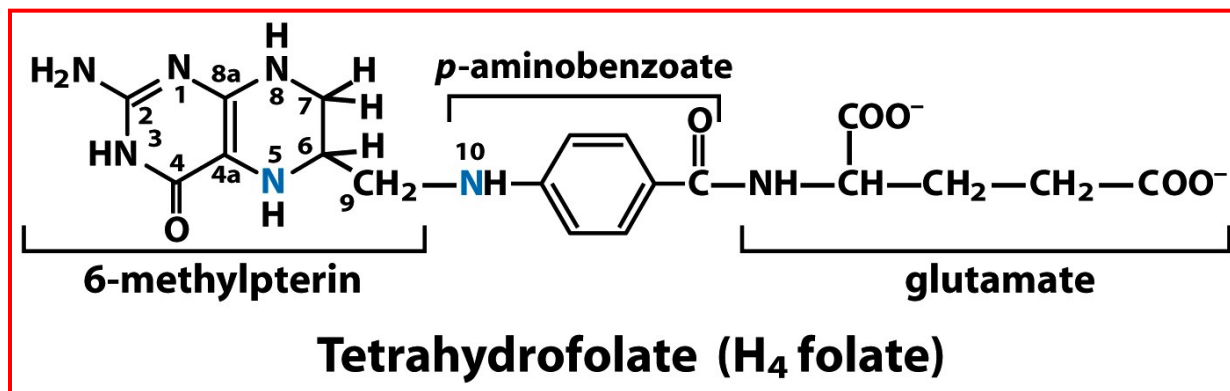
- **Group transfer rxn: coenzyme B12**

Cofactor	Vitamin	Additional component	Chemical group(s) transferred
Thiamine pyrophosphate	Thiamine (B ₁)	pyrophosphate	2-carbon groups, α cleavage
NAD⁺ and NADP⁺	Niacin (B ₃)	ADP	Electrons
Pyridoxal phosphate	Pyridoxine (B ₆)	None	Amino and carboxyl groups
Methylcobalamin	Vitamin B₁₂	Methyl group	acyl groups
Cobalamine	Cobalamine (B ₁₂)	None	hydrogen , alkyl groups
Biotin	Biotin (H)	None	CO ₂
Coenzyme A	Pantothenic acid (B ₅)	ADP	Acetyl group and other acyl groups
Tetrahydrofolic acid	Folic acid (B ₉)	Glutamate residues	Methyl , formyl , methylene and formimino groups
Menaquinone	Vitamin K	None	Carbonyl group and electrons
Ascorbic acid	Vitamin C	None	Electrons
Flavin mononucleotide	Riboflavin (B ₂)	None	Electrons
Flavin adenine dinucleotide	Riboflavin (B ₂)	ADP	Electrons
Coenzyme F420	Riboflavin (B ₂)	Amino acids	Electrons

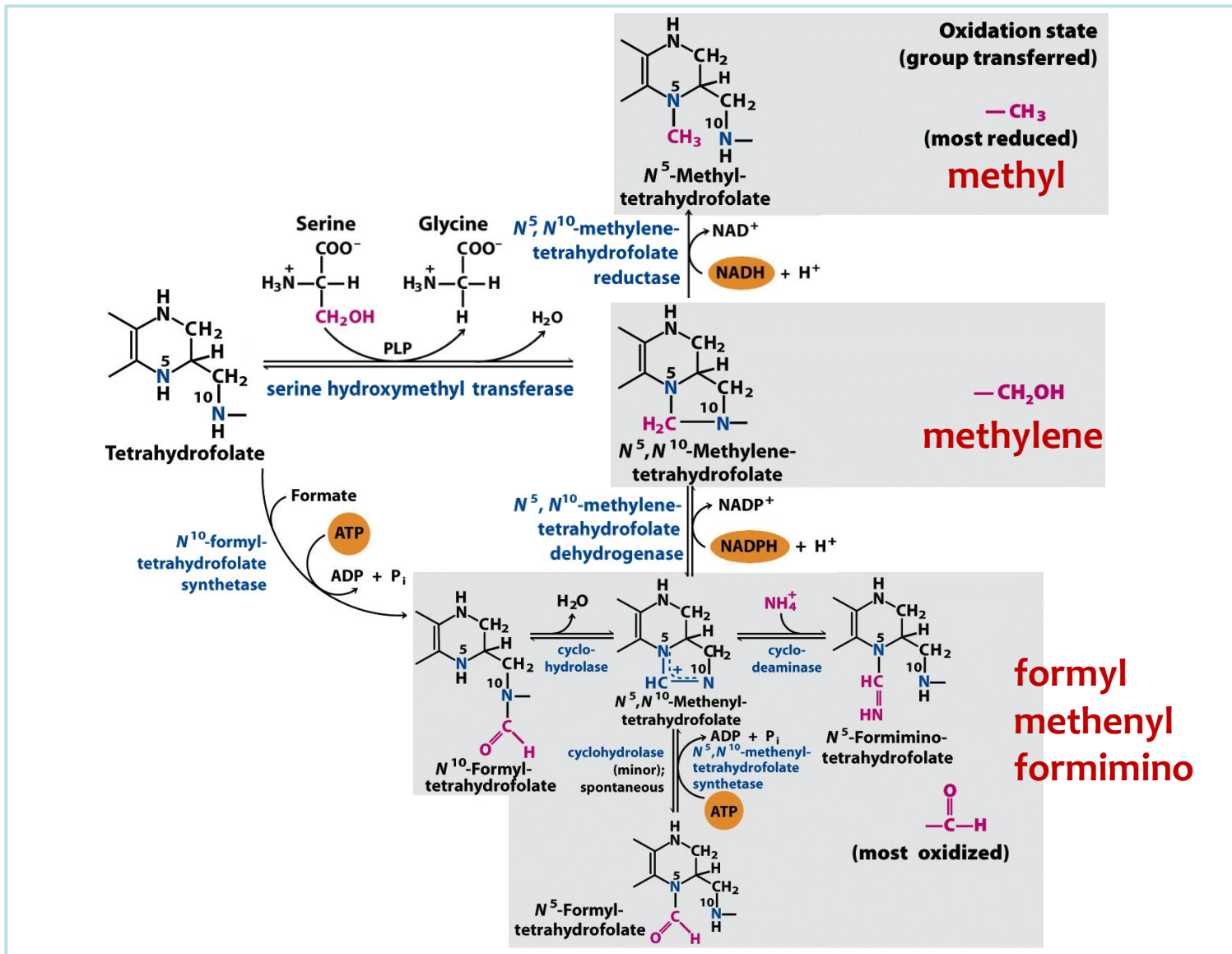
Pyridoxal phosphate for transamination



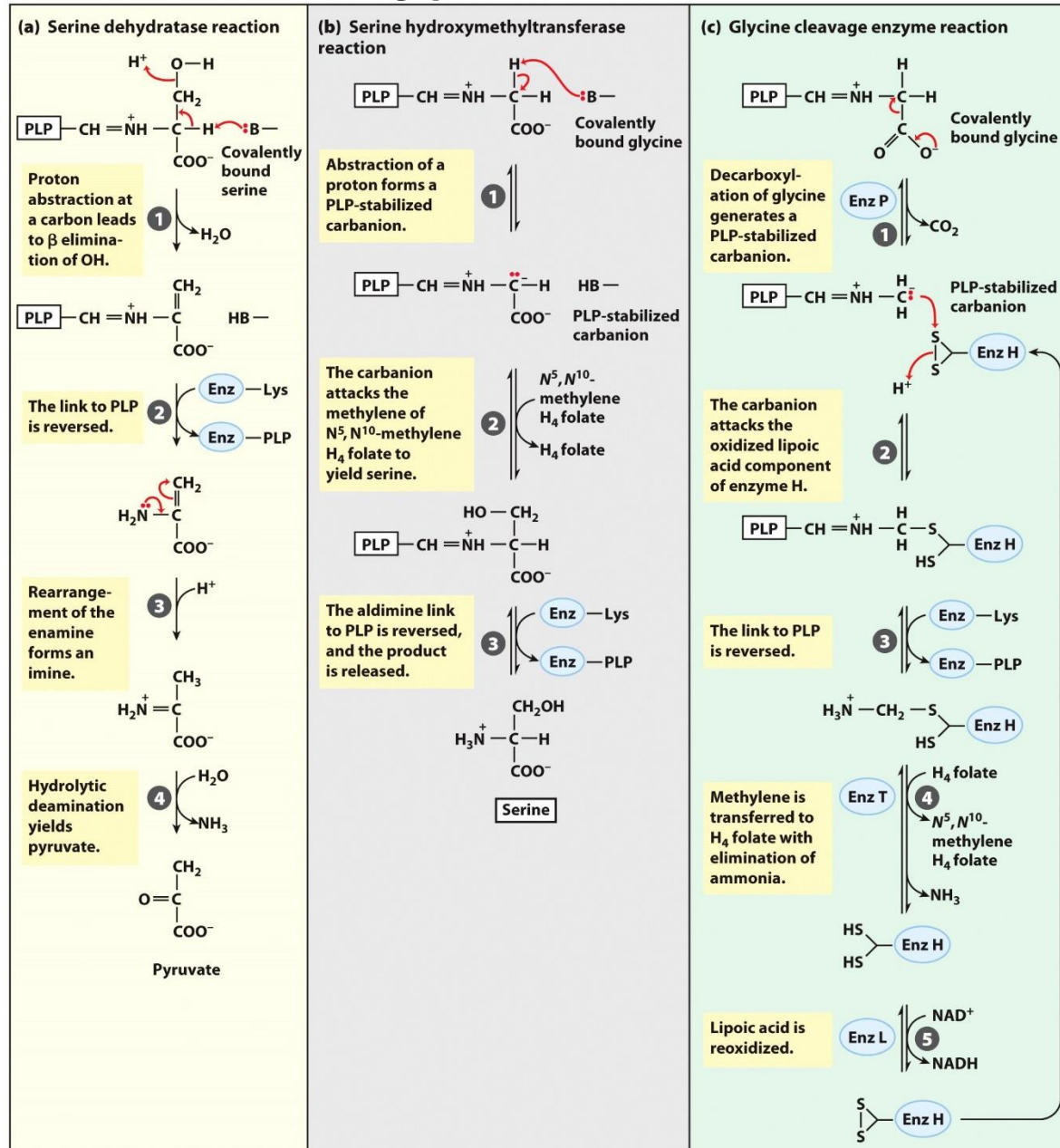
Cofactors for 1C group transfer



Tetrahydrofolate (conversion of 1C units on TFH)

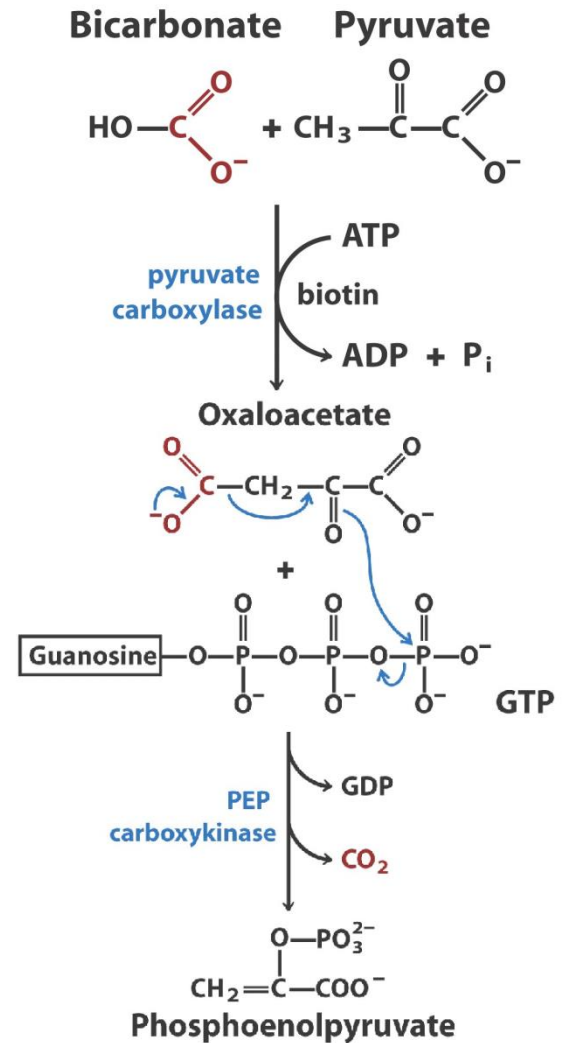
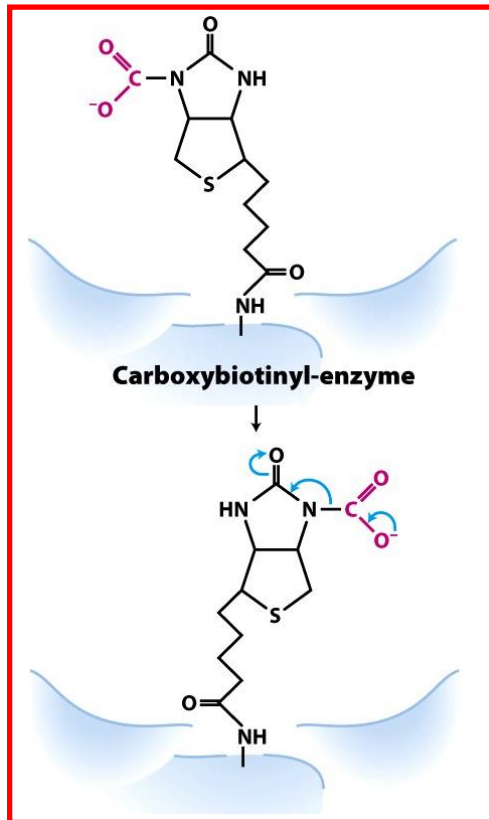


Interplay of the pyridoxal phosphate and tetrahydrofolate cofactors in serine and glycine metabolism

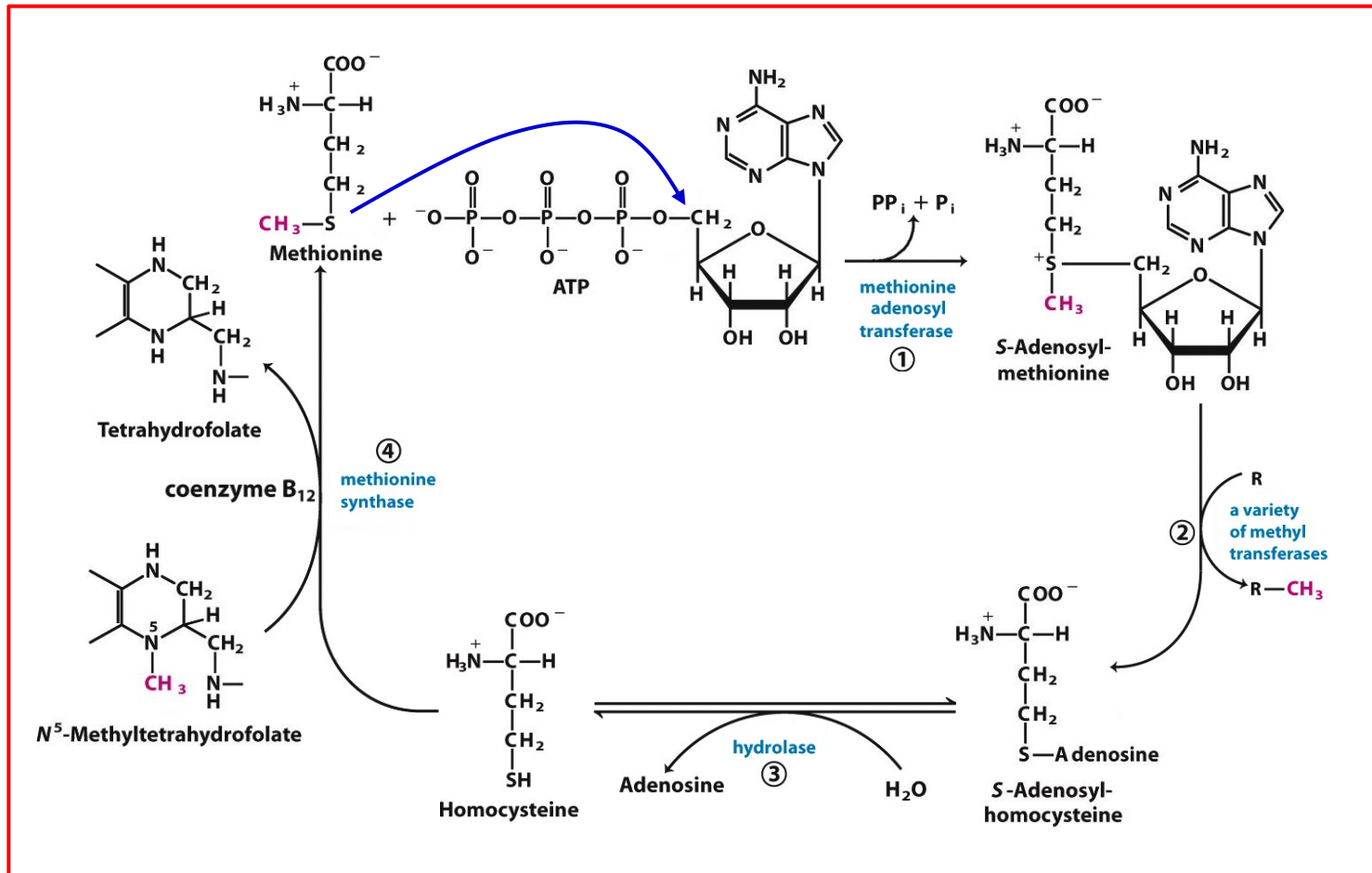


Biotin in pyruvate carboxylase

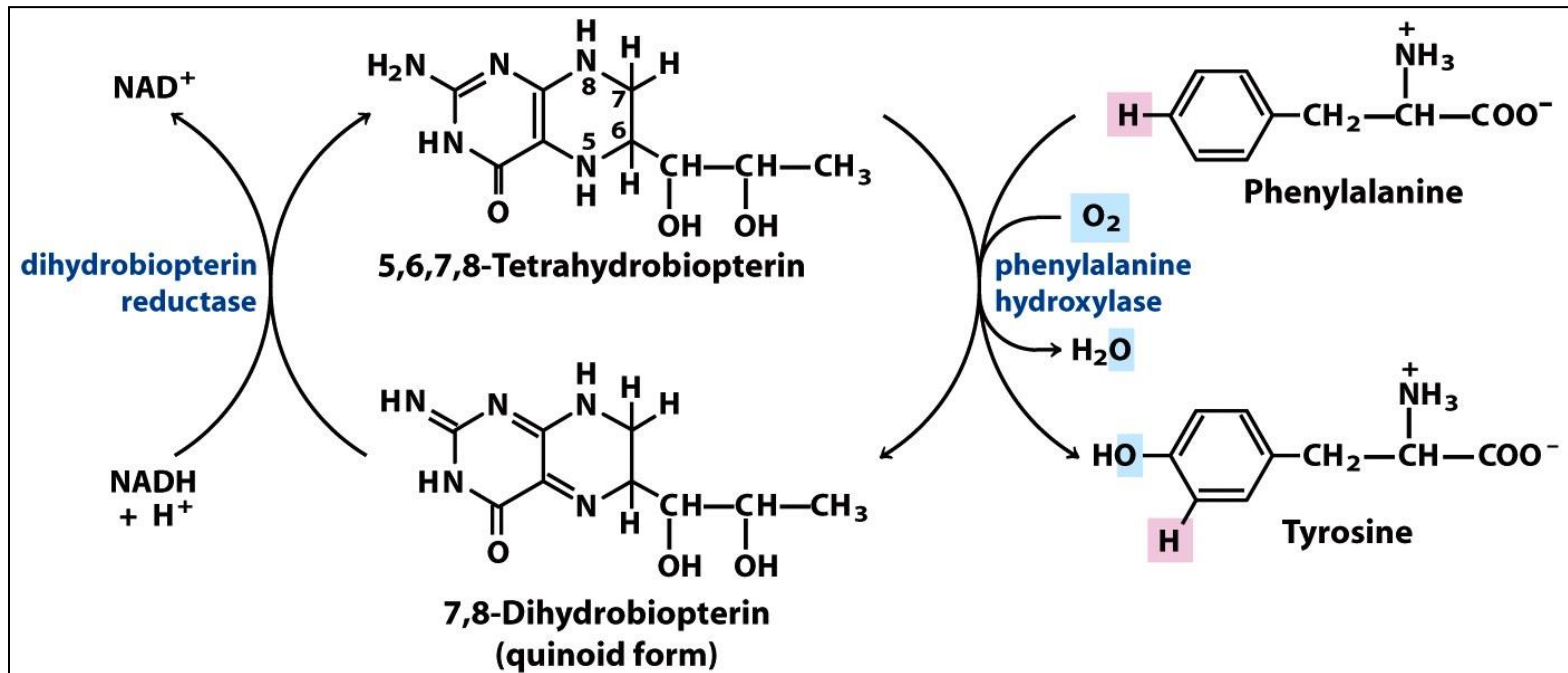
- Bicarbonate activated by ATP
- CO_2 reacts w/ biotin
- CO_2 moves from one site to the other active site



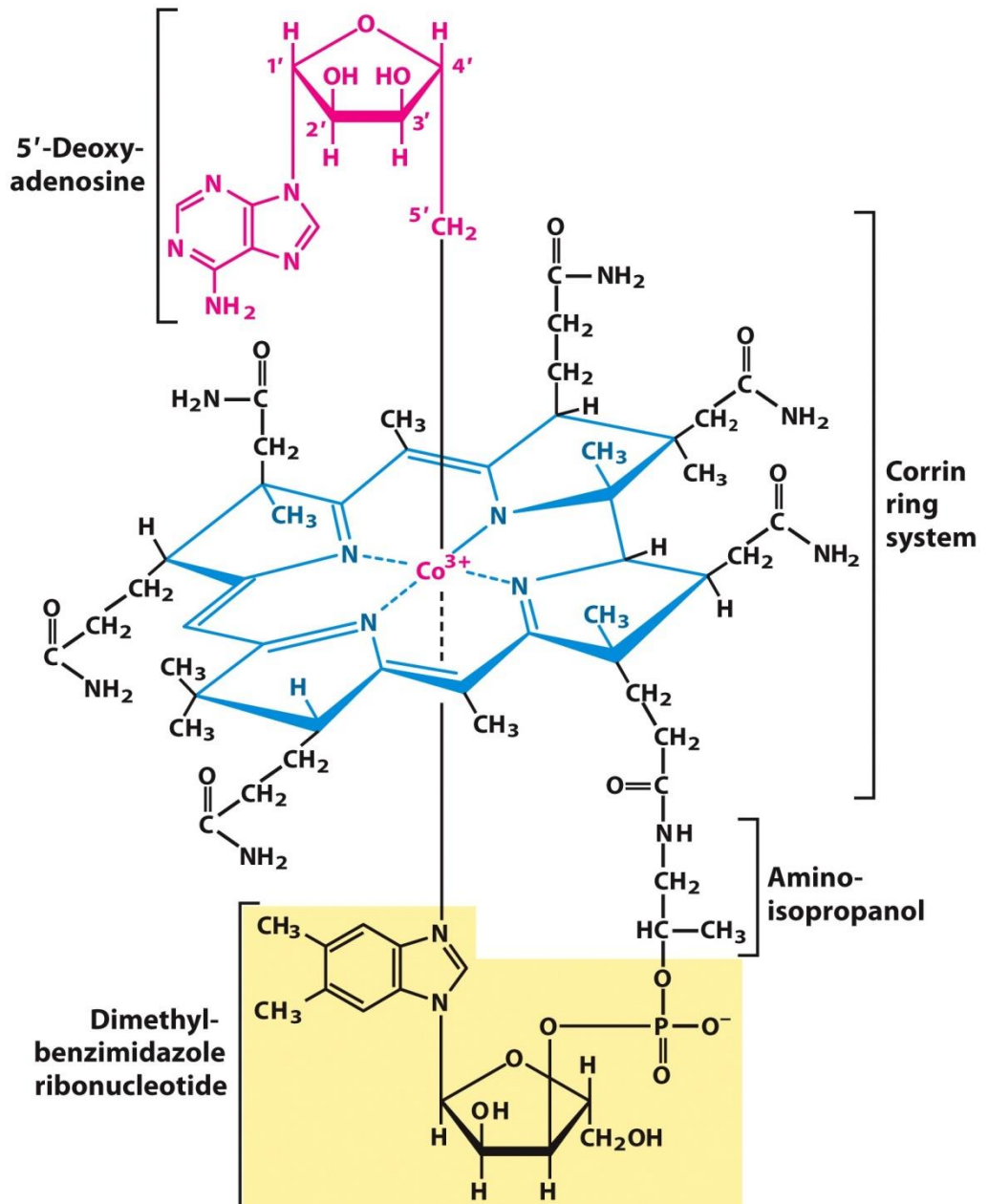
S-adenosylmethionine

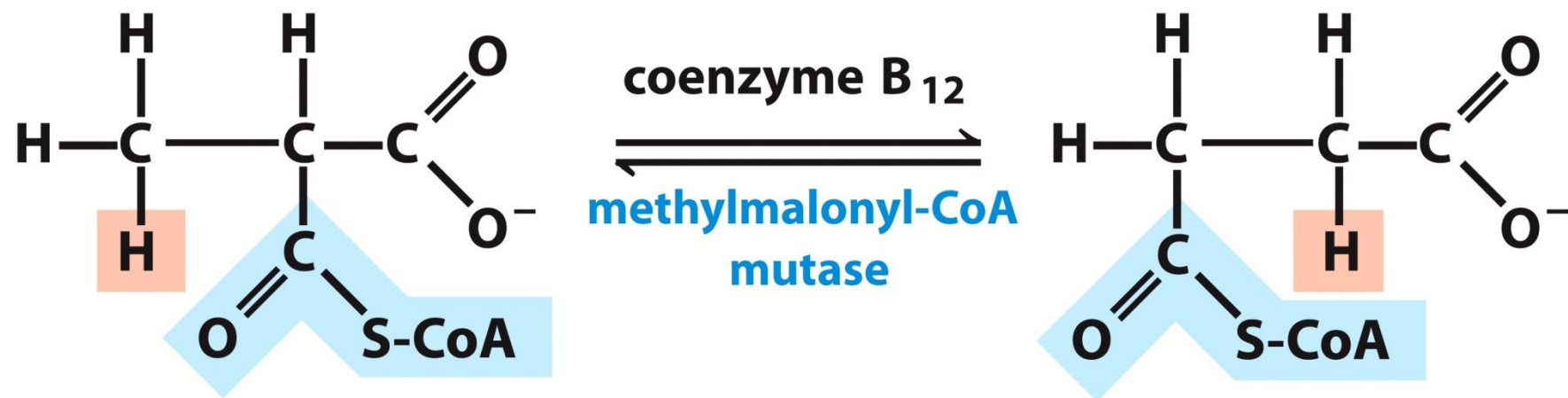


Tetrahydrobiopterin (THBP) : oxidation reaction



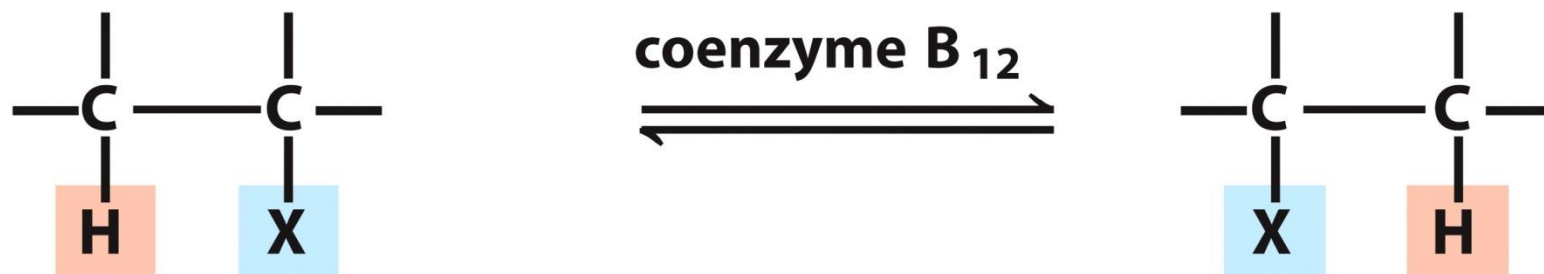
Coenzyme B12 for group transfer





L-Methylmalonyl-CoA

Succinyl-CoA

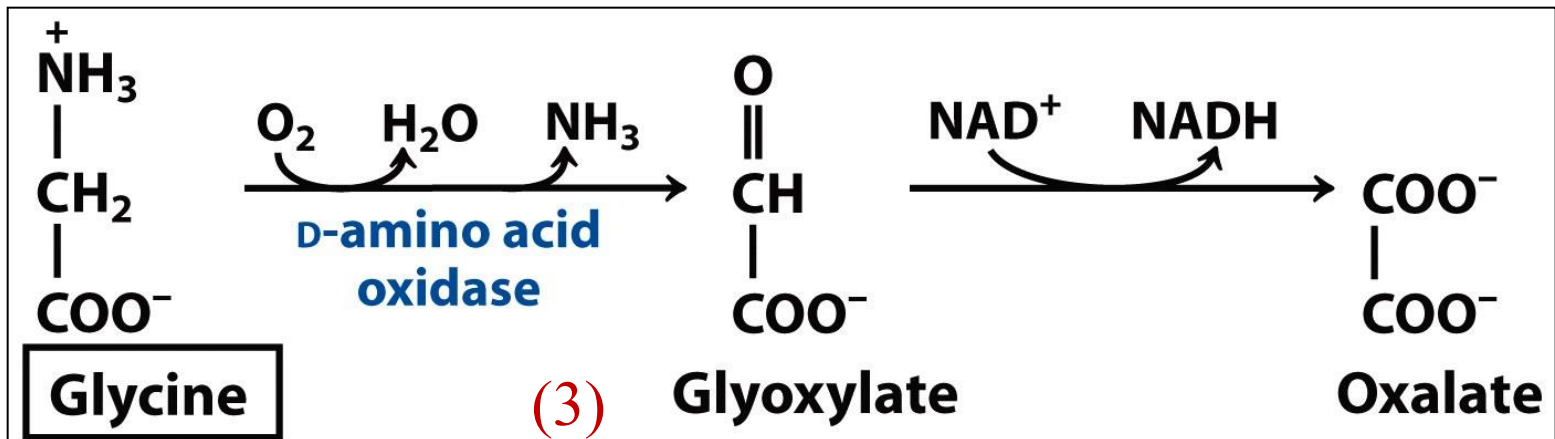
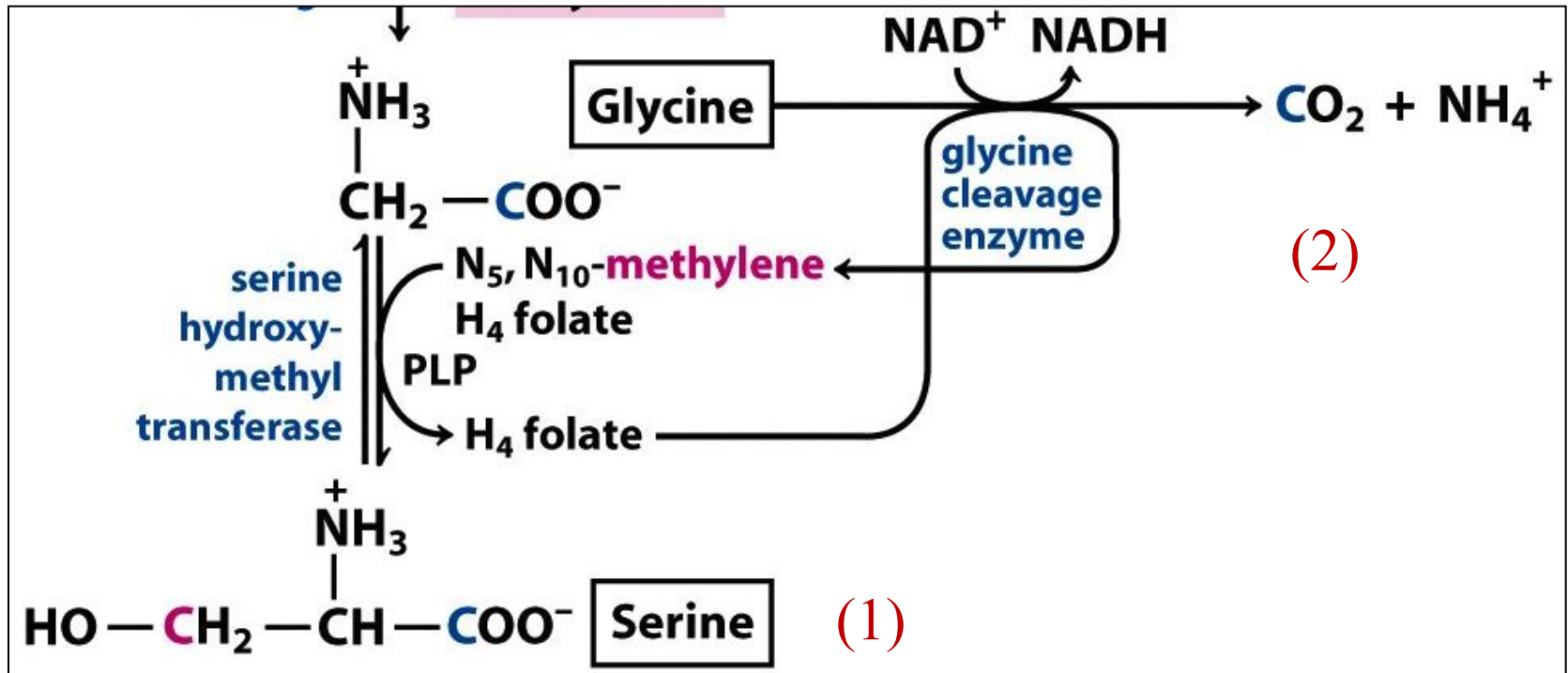


Box 17-2 figure 1

Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

If there are genetic defects...

Glycine metabolism



Defects in glycine cleavage enzyme

: Nonketotic **hyperglycinemia**

: severe mental deficiencies, early death

Hyperglycemia Symptoms



Feeling hungry



Blurry vision



Frequent urination



High blood sugar



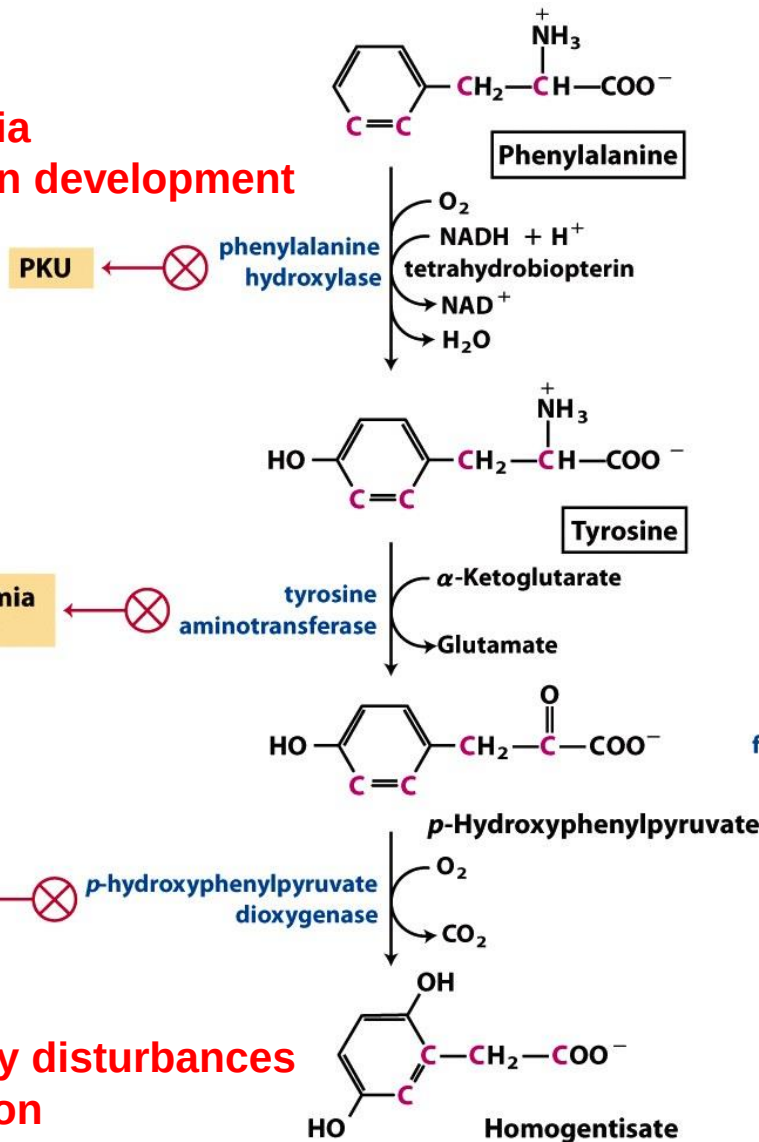
Tingling limbs



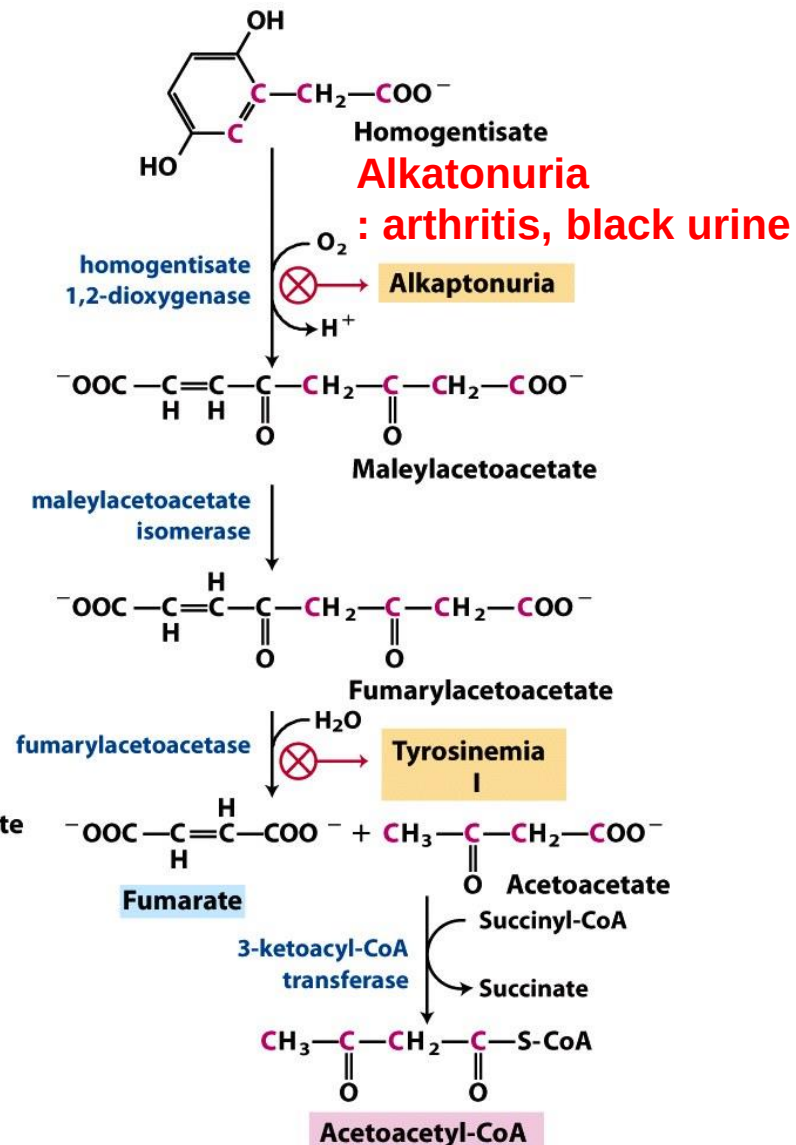
Feeling thirsty

Catabolic pathways for phenylalanine and tyrosine

Phenylketonuria
: impaired brain development



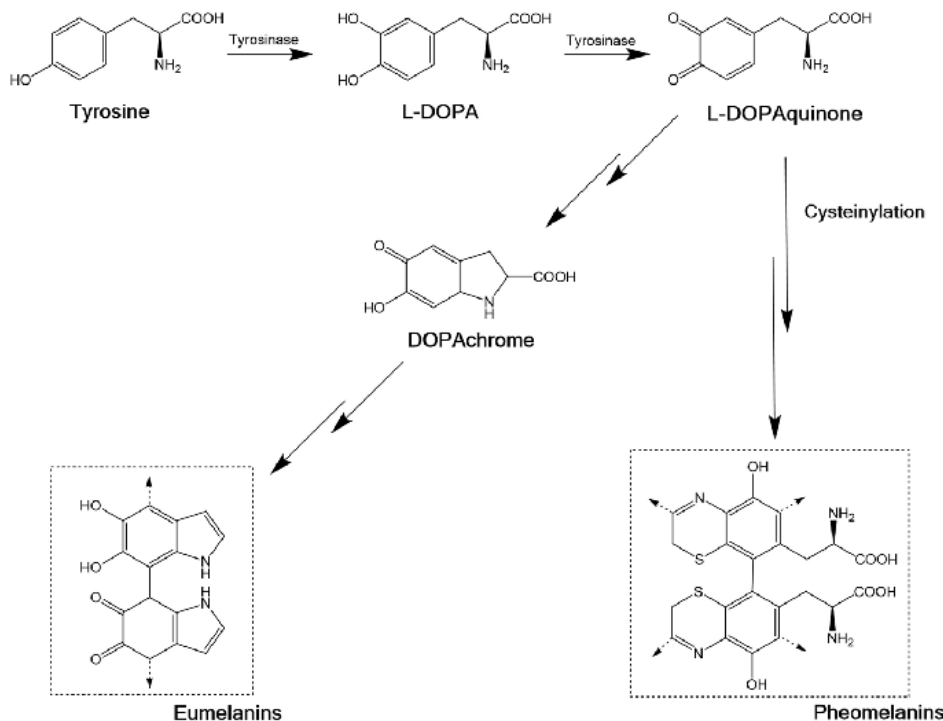
Tyrosinemia
: liver and kidney disturbances
mental retardation



Albinism

('22 약시)

the congenital absence of **melanin** in an animal or plant resulting in white hair, feathers, scales and skin and reddish pink or blue eyes (defective in **tyrosinase**)



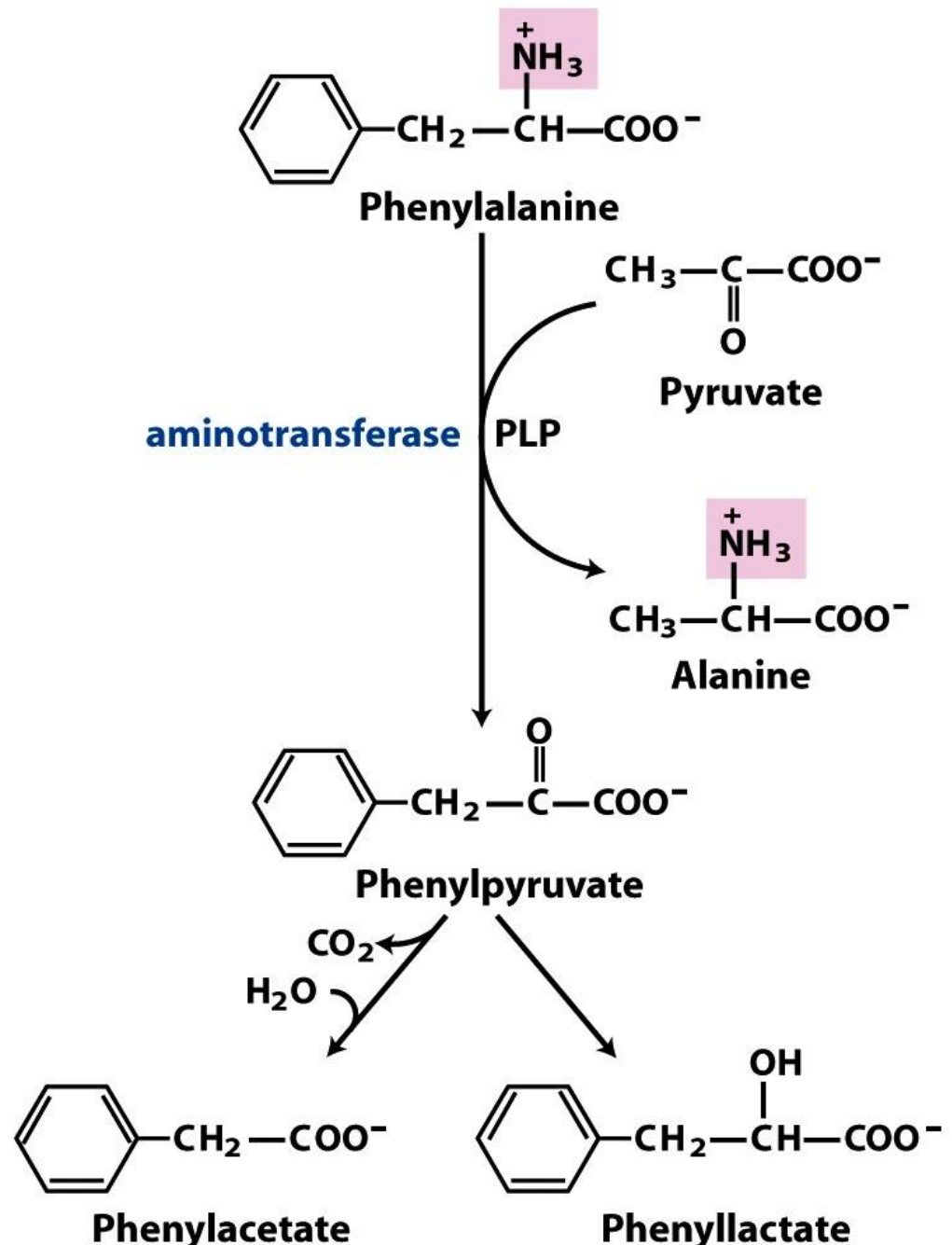
<https://dermmedica.ca/what-is-albinism/>

Alternative pathways for catabolism of phenylalanine in phenylketonuria

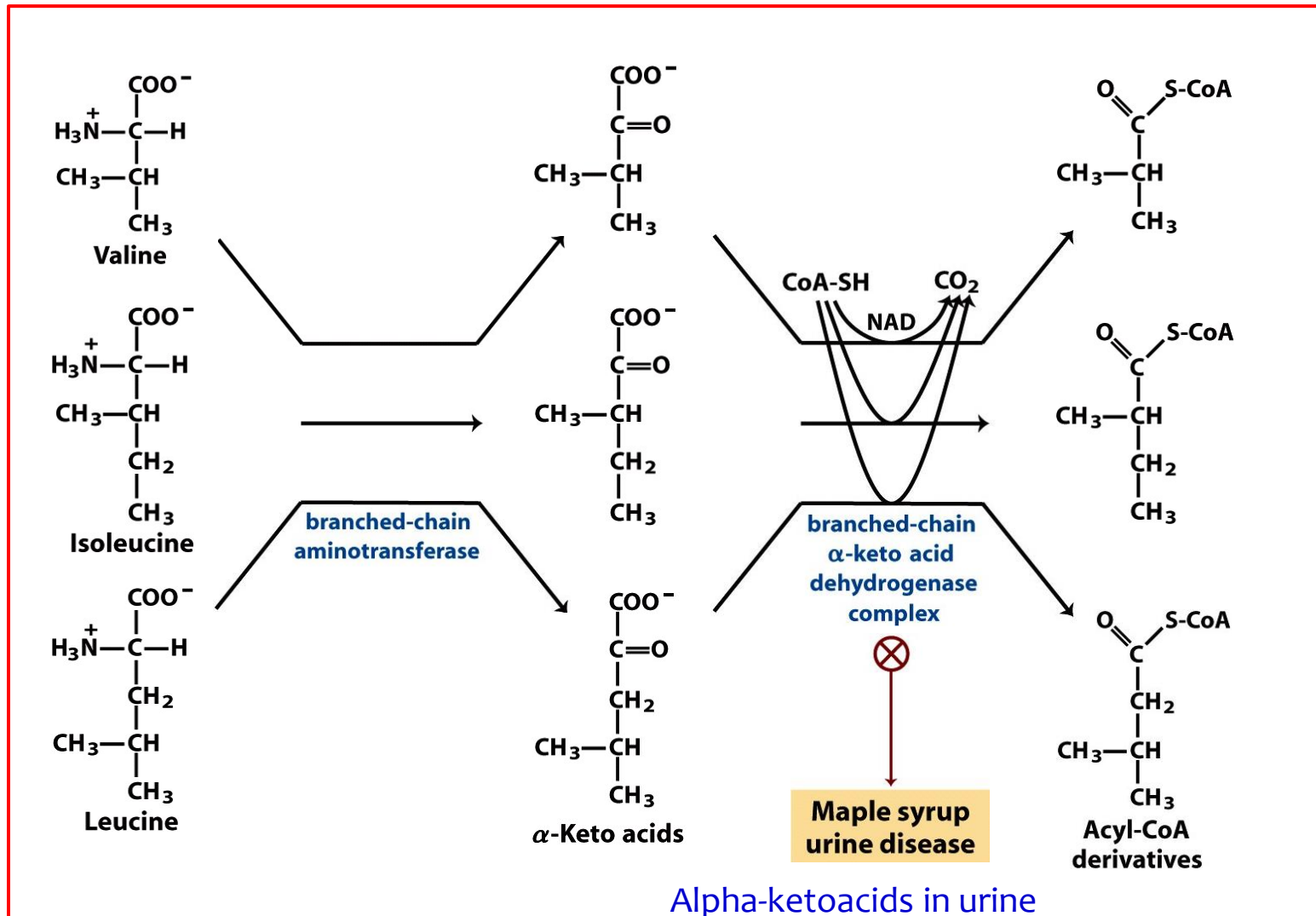
Phenylpyruvate, phenyl acetate, phenyllactate are accumulated in blood and urine

→ Phenylketonuria

→ Impaired brain dev.

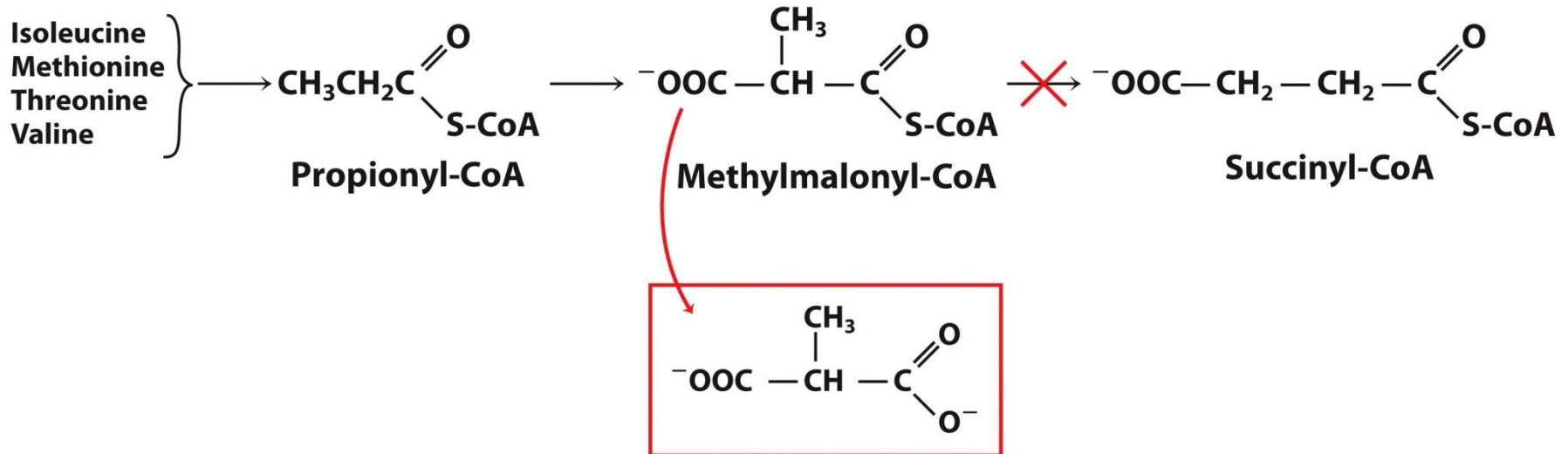


Branched-chain amino acids (VIL)



Alpha-ketoacids in urine
Abnormal brain development
Mental retardation

Methylmalonyl acidemia : defect in methylmalonyl-CoA mutase or vitamin B12 metabolism



**Methylmalonic acid accumulates
in tissues, blood, urine**



Methylmalonic Acidemia

Some Of The Symptoms Of Methylmalonic Acidemia Are:

- 1.) Progressive encephalopathy
- 2.) Severe dehydration
- 3.) Developmental problems
- 4.) Failure to thrive
- 5.) Extreme lethargy
- 6.) Repeated infections
- 7.) Seizures
- 8.) Vomiting.

ePainAssist.com

ePainAssist.com

Pernicious anemia

Vitamin B12 deficiency

- Accumulation of N5-methyl-THF
- Depletion of N5,N10-methylene-THF
- Defect in erythrocyte (RBC) development
- **Megaloblast** (bone marrow immature precursor cell) & **macrocytes** (abnormally large erythrocyte)

Symptoms and Signs of Pernicious Anemia



Weakness



Bleeding of gums



Jaundice



Loss of appetite

TABLE 18–2**Some Human Genetic Disorders Affecting Amino Acid Catabolism**

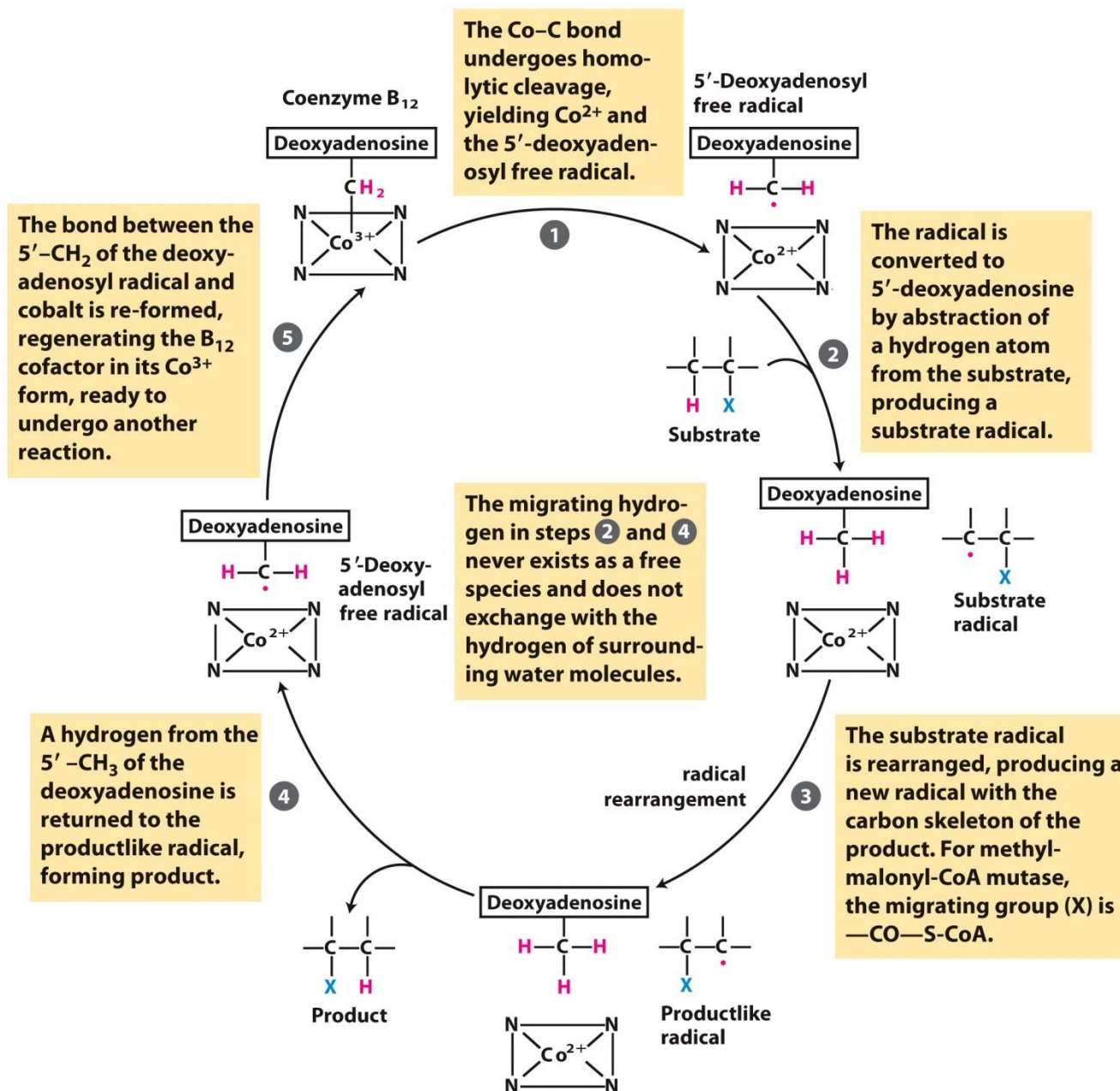
Medical condition	Approximate incidence (per 100,000 births)	Defective process	Defective enzyme	Symptoms and effects
Albinism	<3	Melanin synthesis from tyrosine	Tyrosine 3-monooxygenase (tyrosinase)	Lack of pigmentation; white hair, pink skin
Alkaptonuria	<0.4	Tyrosine degradation	Homogentisate 1,2-dioxygenase	Dark pigment in urine; late-developing arthritis
Argininemia	<0.5	Urea synthesis	Arginase	Mental retardation
Argininosuccinic acidemia	<1.5	Urea synthesis	Argininosuccinase	Vomiting; convulsions
Carbamoyl phosphate synthetase 1 deficiency	<0.5	Urea synthesis	Carbamoyl phosphate synthetase 1	Lethargy; convulsions; early death
Homocystinuria	<0.5	Methionine degradation	Cystathionine β -synthase	Faulty bone development; mental retardation
Maple syrup urine disease (branched-chain ketoaciduria)	<0.4	Isoleucine, leucine, and valine degradation	Branched-chain α -keto acid dehydrogenase complex	Vomiting; convulsions; mental retardation; early death
Methylmalonic acidemia	<0.5	Conversion of propionyl-CoA to succinyl-CoA	Methylmalonyl-CoA mutase	Vomiting; convulsions; mental retardation; early death
Phenylketonuria	<8	Conversion of phenylalanine to tyrosine	Phenylalanine hydroxylase	Neonatal vomiting; mental retardation

Table 18-2*Lehninger Principles of Biochemistry, Fifth Edition*

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Summary

1. Some amino acids are converted to glucose, others to ketone bodies
2. Several enzyme cofactors play important roles in amino acid catabolism
3. 6 amino acids are degraded to pyruvate
4. 7 amino acids are degraded to acetyl-CoA
5. Phe catabolism
6. 5 amino acids are converted to α -ketoglutarate
7. 4 amino acids are converted to succinyl-CoA
8. Branched aa are oxidized in muscle, adipose, kidney and brain
9. Asn and Asp are degraded to oxaloacetate



Box 17-2 figure 4

Lehninger Principles of Biochemistry, Sixth Edition

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